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Tricks with Hicks: The EASI Demand System

By ARTHUR LEWBEL AND KRISHNA PENDAKUR*

We invent Implicit Marshallian demands, which combine desirable features of Hicksian and Marshallian demands. We propose and estimate the Exact Affine Stone Index (EASI) implicit Marshallian demand system. Like the Almost Ideal Demand (AID) system, EASI budget shares are linear in parameters given real expenditures. However, unlike the AID, EASI demands can have any rank and its Engel curves can have any shape over real expenditures. EASI error terms equal random utility parameters to account for unobserved preference heterogeneity. EASI demand functions can be estimated using GMM or three stage least squares, and, like AID, an approximate EASI model can be estimated by linear regression. (JEL D11, D12)

Recent empirical work with large consumer expenditure datasets finds Engel curves (income expansion paths) with significant curvature and variation across goods. For example, some goods have Engel curves that are close to linear or quadratic, while others are more S-shaped (see Richard Blundell, Xiaohong Chen, and Denis Kristensen 2007). Typical parametric demand models cannot encompass this variety of shapes, and are constrained by William Gorman–type rank restrictions (1981).

Other current research shows the importance of allowing for unobserved preference heterogeneity in demand systems, and the difficulty of doing so in a coherent fashion. In most empirical models of consumer demand, model error terms cannot be interpreted as random utility parameters representing unobserved heterogeneity. See, for example, Bryan Brown and Mary Beth Walker (1989), Daniel McFadden and Marcel Richter (1990), Donald Brown and Rosa Matzkin (1998), Lewbel (2001), and Walter Beckert and Blundell (2004).

Despite these empirical issues, Angus Deaton and John Muellbauer's (1980) Almost Ideal Demand (AID) model, which has linear Engel curves and does not incorporate unobserved heterogeneity, remains very popular. This popularity is at least partly because alternative models involve nonlinear functions of many prices and parameters, which are often numerically difficult or intractable to implement. In addition, the AID model has a very convenient approximate form which may be estimated by linear methods.

In this paper, we develop an approach to the specification and estimation of consumer demands that addresses the issues above while maintaining the simplicity of the AID model. Consider a consumer with demographic (and other observable preference related) characteristics $\mathbf{z}$ and log nominal total expenditures x who faces the J -vector of log prices $\mathbf{p}$. Assume she chooses a bundle of goods, described by the J -vector of budget shares $\mathbf{w}$, to maximize utility given her linear

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budget constraint. Hicksian demand functions associated with her utility function, which express $\mathbf{w}$ as a function of $\mathbf{p}$, $\mathbf{z}$, and attained utility level u , can easily be specified to have many desirable properties. We show that, under some conditions that permit both random utility parameters and arbitrarily complicated Engel curves, utility u can be expressed as a simple function of the observed variables x , $\mathbf{w}$, $\mathbf{p}$, and $\mathbf{z}$. This function, which we denote y , can often be interpreted as a measure of log real expenditures. We use these results to directly specify and estimate what we call *implicit Marshallian demands*, which are Hicksian demands after replacing u with the implicit utility measure y .

Noting that $\mathbf{p}'\mathbf{w}$ is the definition of the Stone log price index (see Richard Stone 1954), we define the *Exact Affine Stone Index* (EASI) class of cost functions, which have y equal to an affine transform of Stone index deflated log nominal expenditures, $x - \mathbf{p}'\mathbf{w}$. The resulting EASI implicit Marshallian demand functions have the following properties:

1. Like the AID system, EASI budget share demand functions are, apart from the construction of y , completely linear in parameters, which facilitates estimation in models with many goods.
2. The AID budget shares are linear in $\mathbf{p}$, $\mathbf{z}$, and y . In contrast, EASI budget shares are linear in $\mathbf{p}$ and are polynomials of any order in $\mathbf{z}$ and y . They can also include interaction terms such as $\mathbf{p}y$, $\mathbf{z}y$, and $\mathbf{p}\mathbf{z}'$, and contain other functions of $\mathbf{z}$ and y .
3. EASI Engel curves for each good are almost completely unrestricted. For example, EASI demands can be high-order polynomials or splines in y and $\mathbf{z}$, and so can encompass empirically important specifications that most parametric models cannot capture, such as the semi-parametric S shaped Engel curves reported by Pendakur (1999) and Blundell, Chen, and Kristensen (2007). The AID system is linear in y and has rank two, and the quadratic AID of James Banks, Blundell, and Lewbel (1997) is quadratic in (an approximation of) y with rank three. In contrast, EASI demands can be polynomials or splines of any order in y , and can have any rank up to $J-1$, where J is the number of goods. EASI demands are not subject to the Gorman (1981) rank three limit even with polynomial Engel curves (see the online Appendix, available at <http://www.aeaweb.org/articles.php?doi=10.1257/aer.99.3.827>, for definitions and details regarding the meaning and nature of rank restrictions).
4. EASI budget share error terms can equal unobserved preference heterogeneity or random utility parameters. The AID and other similar models do not have this property, since in those models unobserved preference heterogeneity requires additive errors to be heteroskedastic (see Brown and Walker 1989; Lewbel 2001). The EASI unobserved preference heterogeneity is coherent and invertible (see the online Appendix).
5. EASI demand functions can be estimated using nonlinear instrumental variables, particularly nonlinear three-stage least squares (3SLS) and Lars Hansen's (1982) Generalized Method of Moments (GMM). Like the AID system, approximate versions of EASI demands can be estimated by linear regression. We find little empirical difference between exact nonlinear and approximate linear EASI estimates.
6. Since EASI demands are derived from a cost function model, given estimated parameters we have simple closed form expressions for consumer surplus calculations, such as cost-of-living indices for large price changes.

EASI demands thus accommodate an extremely wide class of functional forms. In the empirical work below, we implement a model where, for the budget share w_j of each good j , the estimating equations for demand systems have the linear-in-parameters form

$$w_j = \sum_{r=0}^5 b_{rj} y^r + \sum_{l=1}^L (C_{lj} z_l + D_{lj} z_l y) + \sum_{l=0}^L \sum_{k=1}^J A_{lkj} z_l p_k + \sum_{k=1}^J B_{kj} p_k y + \varepsilon_j,$$

where y is a measure of real total expenditures. The regressors in this model are a fifth-order polynomial in y , log-prices p_k of each good k , and L different demographic characteristics z_l , in addition to interaction terms of the forms $p_k y$, $z_l y$, and $z_l p_k$. Our approximate EASI model estimates these equations for each good j by ordinary least squares, letting y equal $x - \sum_{j=1}^J p_j w_j$, while the exact model has y given by equation (8) below.

We begin with an overview of our approach, defining implicit Marshallian demand functions and general EASI demands. We show how these models accommodate high rank engel curves and unobserved preference heterogeneity. We describe the specific EASI functional form we will use in our empirical application, and also show how to construct a fully linear approximation for our EASI demand model, analogous to the approximate linear AID model. We show how to apply the EASI model with consumer surplus and compensated elasticity calculations. Then we describe estimators for the EASI model, including consistent, asymptotically normal instrumental variable, 3SLS, and GMM estimators for the exact model and linear least squares estimators for the approximate model.

We estimate the exact model using Canadian micro-data. We find more complicated Engel curve shapes than those of standard parametric demand systems, and we find that the simple approximate EASI model captures this complexity very well. We apply the model to a cost-of-living experiment, and find that both the increased flexibility of Engel curves and the incorporation of unobserved heterogeneity into the model significantly affect the resulting welfare calculations.

The Appendix provides formal theorems and proofs, along with some extensions and additional mathematical properties of EASI and other implicit Marshallian demands, which are relevant for evaluating these models and for other possible applications of our general methodology (see Appendix A, B, and C in this article and further materials available online, including a supplemental Appendix).

I. Methods and Models

Here we define the general idea of implicit Marshallian demands, describe the EASI functional form, and explain how this model can be estimated and applied. Formal theorems regarding these methods and models are provided in the Appendix and online Appendix.

A. Implicit Marshallian Demand Functions

We specify a cost (expenditure) function and use Shephard's lemma to obtain Hicksian demands that have the desired properties. The usual next step would be to obtain Marshallian demands, which are functions of $\mathbf{p}$, $\mathbf{z}$, and x , by solving for indirect utility u in terms of $\mathbf{p}$, $\mathbf{z}$, and x , and substituting this into Hicksian demand functions. We instead construct cost functions that have simple expressions for u in terms of $\mathbf{w}$, $\mathbf{p}$, $\mathbf{z}$, and x . We call this expression, y , implicit utility and substitute y for u in the Hicksian demands to yield what we call implicit Marshallian demand functions. These implicit Marshallian demands circumvent the difficulty of finding simple analytic expressions for indirect utility or Marshallian demands.

We wish to explicitly include both observable and unobservable sources of preference heterogeneity in our models, so let $\mathbf{z} = (z_1, \dots, z_L)'$ be an L -vector of observable demographic (or

other) characteristics that affect preferences, and let ϵ be a J -vector of unobserved preference characteristics (taste parameters) satisfying $\mathbf{1}'\epsilon = 0$, where $\mathbf{1}_J$ is the J -vector of ones. Typical elements of $\mathbf{z}$ would include household size, age, and composition. The log cost or expenditure function is $x = C(\mathbf{p}, u, \mathbf{z}, \epsilon)$, which equals the minimum log-expenditure required for an individual with characteristics $\mathbf{z}, \epsilon$ to attain utility level u when facing log prices $\mathbf{p}$. Shephard's Lemma expresses Hicksian (compensated) budget-share functions, $\omega(\mathbf{p}, u, \mathbf{z}, \epsilon)$, as $\mathbf{w} = \omega(\mathbf{p}, u, \mathbf{z}, \epsilon) = \nabla_{\mathbf{p}} C(\mathbf{p}, u, \mathbf{z}, \epsilon)$. Indirect utility, V , is the inverse of log-cost with respect to u : $u = V(\mathbf{p}, x, \mathbf{z}, \epsilon) = C^{-1}(\mathbf{p}, \cdot, \mathbf{z}, \epsilon)$. If an analytic solution for V is unavailable, it may still be possible to express utility as a function g of $\mathbf{w}, \mathbf{p}, x, \mathbf{z}$. In this case, we can write $u = g(\omega(\mathbf{p}, u, \mathbf{z}, \epsilon), \mathbf{p}, x, \mathbf{z})$, which implicitly defines u . We may then define *implicit utility* y by $y = g(\mathbf{w}, \mathbf{p}, x, \mathbf{z})$. By construction, $y = u$ and the function g that defines y depends only on observable data, and may have a simple tractable closed-form expression even when no closed-form expression exists for indirect utility V . The *implicit Marshallian demand* system is then given by $\mathbf{w} = \omega(\mathbf{p}, y, \mathbf{z}, \epsilon)$, which is the Hicksian demand system except u is replaced with y .

A similar idea to implicit Marshallian demands is Martin Browning's (2001) "M-demands," which expresses demand functions in terms of prices and the quantity of one good, instead of in terms of prices and total expenditures. This can be interpreted as a restrictive choice of g , since it uses only one good.

In this construction, preferences, and hence utility, remain ordinal. So when we say that implicit utility y equals the utility level u , this means only that the observable y equals one possible cardinalization of u . In all our examples y will be linear in x and will either equal or closely approximate a money metric cardinalization of utility, and hence y will also be interpretable as a measure of real expenditures. We will sometimes refer to y as log real expenditures when this interpretation of implicit utility is particularly relevant.

Since g expresses utility as a function of all the arguments of indirect utility plus allowing for dependence ω , it admits more possibilities than are available for explicit Marshallian demand systems. Although, in theory, the idea of implicit demand systems opens up an extremely large class of potential demand systems, we find that simple forms for g allow us to solve a large number of problems facing empirical demand analysis.

To illustrate the idea of implicit Marshallian demands, consider the following restrictive example. For simplicity (ignoring the inequality constraints required for demand system regularity), consider a log cost function of the form

$$(1) \quad C(\mathbf{p}, u, \mathbf{z}, \epsilon) = u + \mathbf{p}'\mathbf{m}(u, \mathbf{z}) + \mathbf{p}'\epsilon,$$

where $\mathbf{m}(u, \mathbf{z})$ is a J -vector valued function with $\mathbf{1}'\mathbf{m}(u, \mathbf{z}) = 1$. By Shephard's lemma, this cost function has Hicksian (compensated) budget shares $\mathbf{w} = \omega(\mathbf{p}, u, \mathbf{z}, \epsilon) = \mathbf{m}(u, \mathbf{z}) + \epsilon$. Since $x = u + \mathbf{p}'[\mathbf{m}(u, \mathbf{z}) + \epsilon]$, we may substitute $\mathbf{m}(u, \mathbf{z}) + \epsilon$ with $\mathbf{w}$ to derive $u = x - \mathbf{p}'\mathbf{w}$. We therefore obtain implicit Marshallian demands with budget shares of the form

$$\begin{aligned} \mathbf{w} &= \mathbf{m}(x - \mathbf{p}'\mathbf{w}, \mathbf{z}) + \epsilon \\ &= \mathbf{m}(y, \mathbf{z}) + \epsilon, \end{aligned}$$

where $y = g(\mathbf{w}, \mathbf{p}, x, \mathbf{z}) = x - \mathbf{p}'\mathbf{w}$. Here, implicit utility $y = x - \mathbf{p}'\mathbf{w}$ equals the log of nominal expenditures x deflated by the log Stone (1954) price index $\mathbf{p}'\mathbf{w}$. Here, $\exp(y)$ is equal to nominal expenditure deflated by a price index equal to one when all prices equal one. For this reason, we may interpret y in this example as the log of real expenditures.

Unlike the AID system, where a Stone index is used to approximate the correct deflator for x , in this model the Stone index is the exact, correct deflator for obtaining real expenditures. This

is an example of what we call an Exact Stone Index cost function, which is any cost function for which $y = u = x - \mathbf{p}'\mathbf{w}$ (the details of this class are provided in Appendix A). This model does not have a closed-form expression for indirect utility or ordinary Marshallian demand functions, except for very special choices of $\mathbf{m}(u, \mathbf{z})$, but it can still be readily estimated because it expresses budget shares as functions of observables $\mathbf{p}, \mathbf{z}$ and $y = x - \mathbf{p}'\mathbf{w}$. This example also shows how implicit Marshallian demands can easily incorporate unobserved preference heterogeneity, since the error terms in the nonlinear regression of $\mathbf{w}$ on y equal the random utility parameters $\boldsymbol{\varepsilon}$. This example also shows that the functional form of budget shares over $y, \mathbf{z}$ is completely unrestricted, and Gorman's rank three limit does not apply. Finally, although y is endogenous (it depends on $\mathbf{w}$), it can be instrumented (for example, by x and $\mathbf{p}$), so estimation can take the form of instrumental variables, nonlinear 3SLS, or GMM.

This example model is clearly very restrictive, particularly regarding price effects that enter the model only through y . More generally, in Appendix A we show (by Theorems 1 and 2) that all Exact Stone Index demand systems have some undesirable properties, so we instead propose a generalization in which implicit utility equals an affine transform of the log of Stone-index deflated expenditures. These are the Exact Affine Stone Index (EASI) implicit Marshallian demand functions.

Theorems 3 and 4 in Appendix B describe EASI cost functions, which are those with the property that utility is ordinaly equivalent to an affine transformation of $x - \mathbf{p}'\mathbf{w}$. Equation (B5) in Appendix B describes one such cost function that is particularly convenient for empirical implementation:

$$(2) \quad C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = u + \mathbf{p}'\mathbf{m}(u, \mathbf{z}) + T(\mathbf{p}, \mathbf{z}) + S(\mathbf{p}, \mathbf{z})u + \mathbf{p}'\boldsymbol{\varepsilon},$$

This cost function differs from (1) only in the addition of $T(\mathbf{p}, \mathbf{z}) + S(\mathbf{p}, \mathbf{z})u$. It is similar to the class of cost functions $C(\mathbf{p}, u) = u + \mathbf{p}'\mathbf{m}(u) + \frac{1}{2}\mathbf{p}'M(u)\mathbf{p}$ proposed by Krishna Pendakur and Stefan Sperlich (2008), though they estimate ordinary Marshallian demands for their model by numerically solving for u . There is also a connection to Gorman's (1976) "Tricks with Utility Functions," though we employ an affine transform of deflated expenditures instead of affine transforms of price or quantity vectors.

By Theorem 4 in Appendix B, this class of cost functions has implicit Marshallian demands $\mathbf{w} = \boldsymbol{\omega}(\mathbf{p}, y, \mathbf{z}, \boldsymbol{\varepsilon})$ given by

$$(3) \quad \mathbf{w} = \mathbf{m}(y, \mathbf{z}) + \nabla_{\mathbf{p}}T(\mathbf{p}, \mathbf{z}) + \nabla_{\mathbf{p}}S(\mathbf{p}, \mathbf{z})y + \boldsymbol{\varepsilon}$$

where y is given by

$$(4) \quad u = y = \frac{x - \mathbf{p}'\mathbf{w} - T(\mathbf{p}, \mathbf{z}) + \mathbf{p}'[\nabla_{\mathbf{p}}T(\mathbf{p}, \mathbf{z})]}{1 + S(\mathbf{p}, \mathbf{z}) - \mathbf{p}'[\nabla_{\mathbf{p}}S(\mathbf{p}, \mathbf{z})]}$$

which is an affine transform of the log of Stone Index deflated nominal expenditures. Properties of this general class of EASI models are provided in Appendix B.

B. An EASI Demand Model for Estimation

We propose the following parametric EASI cost function as our baseline case for empirical work:

$$(5) \quad C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = u + \mathbf{p}' \left[\sum_{r=0}^5 \mathbf{b}_r u^r + \mathbf{Cz} + \mathbf{Dzu} \right] + \frac{1}{2} \sum_{l=0}^L z_l \mathbf{p}'\mathbf{A}_l \mathbf{p} + \frac{1}{2} \mathbf{p}'\mathbf{Bp}u + \mathbf{p}'\boldsymbol{\varepsilon}.$$

Here, $z_0 = 1$ and (for notational convenience inside the $\mathbf{m}$ function) z_0 is not an element of the vector $\mathbf{z} = (z_1, \dots, z_L)'$; each $\mathbf{b}_r$ is a J -vector of parameters with $\mathbf{1}'_J \mathbf{b}_0 = 1$, $\mathbf{1}'_J \mathbf{b}_r = 0$ for $r \neq 0$; $\mathbf{A}_l$ for $l = 0, \dots, L$, and $\mathbf{B}$ are $J \times J$ symmetric matrices with $\mathbf{1}'_J \mathbf{A}_l = \mathbf{1}'_J \mathbf{B} = \mathbf{0}'_J$; and both $\mathbf{C}$ and $\mathbf{D}$ are $L \times J$ matrices with $\mathbf{1}'_J \mathbf{C} = \mathbf{1}'_J \mathbf{D} = \mathbf{0}_L$.

This model for empirical application is the special case of the EASI model (2) in which

$$T(\mathbf{p}, \mathbf{z}) = \frac{1}{2} \sum_{l=0}^L z_l \mathbf{p}' \mathbf{A}_l \mathbf{p},$$

$$S(\mathbf{p}, \mathbf{z}) = \frac{1}{2} \mathbf{p}' \mathbf{B} \mathbf{p},$$

and

$$(6) \quad \mathbf{m}(u, \mathbf{z}) = \left(\sum_{r=0}^5 \mathbf{b}_r u^r \right) + \mathbf{C} \mathbf{z} + \mathbf{D} \mathbf{z} u.$$

The vector of functions $\mathbf{m}(u, \mathbf{z})$, which generates the model's Engel curves, could be replaced with other sets of flexible functions of $u, \mathbf{z}$ such as splines. We use a polynomial in u and an affine function of $\mathbf{z}$ to strike a balance between simple tractability and general Engel curve flexibility. We choose quadratic forms in prices for T and S because these have simple gradient vectors $\nabla_{\mathbf{p}} T(\mathbf{p}, \mathbf{z})$ and $\nabla_{\mathbf{p}} S(\mathbf{p}, \mathbf{z})$. Finally, we make S independent of $\mathbf{z}$ for the sake of terseness.

By Shephard's lemma, the cost function (5) has Hicksian (compensated) budget shares

$$(7) \quad \mathbf{w} = \sum_{r=0}^5 \mathbf{b}_r u^r + \mathbf{C} \mathbf{z} + \mathbf{D} \mathbf{z} u + \sum_{l=0}^L z_l \mathbf{A}_l \mathbf{p} + \mathbf{B} \mathbf{p} u + \boldsymbol{\varepsilon}.$$

It can be readily checked from these formulas that

$$C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = u + \mathbf{p}' \mathbf{w} - \sum_{l=0}^L (z_l \mathbf{p}' \mathbf{A}_l \mathbf{p} / 2) - \mathbf{p}' \mathbf{B} \mathbf{p} u / 2,$$

and solving this expression for u yields implicit utility y :

$$(8) \quad y = g(\mathbf{w}, \mathbf{p}, x, \mathbf{z}) = \frac{x - \mathbf{p}' \mathbf{w} + \sum_{l=0}^L z_l \mathbf{p}' \mathbf{A}_l \mathbf{p} / 2}{1 - \mathbf{p}' \mathbf{B} \mathbf{p} / 2}.$$

This y has many of the properties of log real expenditures. It equals a cardinalization of utility u , it is affine in nominal expenditures x , and it equals x in the base period when all prices equal one (which is when log prices $\mathbf{p}$ equal zero). Also, when $\mathbf{B}$ is zero, y exactly equals the log of nominal expenditures deflated by a price index. We later show empirically that y is very highly correlated with the log of stone index deflated nominal expenditures, which is a very popular ad hoc measure of real expenditures. Like any money-metric utility measure, y is just a mathematically convenient representation of utility and need not have any deeper significance as an objective or interpersonally comparable utility measure (see the "Shape Invariance and Equivalence Scales" section of the online Appendix for details).

Substituting implicit utility y into the Hicksian budget shares yields the implicit Marshallian budget shares

$$(9) \quad \mathbf{w} = \sum_{r=0}^5 \mathbf{b}_r y^r + \mathbf{Cz} + \mathbf{Dz}y + \sum_{l=0}^L z_l \mathbf{A}_l \mathbf{p} + \mathbf{Bpy} + \boldsymbol{\epsilon},$$

which is the matrix form of the equation provided in the introduction.

Given y , which is a function of observables x , $\mathbf{p}$, $\mathbf{z}$, and the log Stone index $\mathbf{p}'\mathbf{w}$, the budget share equations (9) are linear in parameters and so can be easily estimated. An approximate estimator applies ordinary (linear) least squares after replacing y with $x - \mathbf{p}'\mathbf{w}$ in (9), just like estimation of the approximate AID model. Better estimators simultaneously estimate the model (9) with the parameters in y given by (8). One could use 3SLS or GMM estimation to account for the endogeneity that results from having $\mathbf{w}$ appear in y . Use of GMM would allow for possible unknown heteroskedasticity in $\boldsymbol{\epsilon}$.

The EASI budget shares (9) have compensated price effects governed by $\mathbf{A}_l$, $l = 0, 1, 2, \dots, L$, and $\mathbf{B}$, which allow for flexible price effects and for flexible interactions of these effects with expenditure and with observable demographic characteristics. The Engel curve terms $\mathbf{b}_r$, $r = 0, 1, 2, \dots, 5$ specify budget shares as fifth-order polynomials in y , which (recalling that y is affine in log nominal expenditures x) allows Engel curves to have very complicated shapes. Some empirically popular demand functions have budget shares quadratic in log total expenditures, corresponding to $r = 0, 1, 2$. We added higher moments $r = 3, 4, 5$ because they were statistically significant. The terms $\mathbf{C}$ and $\mathbf{D}$ allow demographic characteristics to enter budget shares through both intercept and slope terms on y . The random utility parameters $\boldsymbol{\epsilon}$, representing unobserved preference heterogeneity, take the form of simple additive errors in the implicit Marshallian demand equations.

The complete set of properties required for the EASI cost function (5) to satisfy cost function regularity is as follows. The adding up constraint and cost function homogeneity conditions are satisfied with $\mathbf{1}'_j \mathbf{b}_0 = 1$, $\mathbf{1}'_j \mathbf{b}_r = 0$ for $r \neq 0$, $\mathbf{1}'_j \mathbf{A}_l = \mathbf{1}'_j \mathbf{B} = \mathbf{0}'_j$, $\mathbf{1}'_j \mathbf{C} = \mathbf{1}'_j \mathbf{D} = \mathbf{0}'_j$, and $\boldsymbol{\epsilon}'_j \mathbf{1}_j = 0$. Symmetry of $\mathbf{A}_l$ and $\mathbf{B}$ ensures Slutsky symmetry. Strict monotonicity of cost requires $\partial C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\epsilon}) / \partial u > 0$, which implies $\mathbf{p}'[(\sum_{r=0}^5 \mathbf{b}_r r u^{r-1}) + \mathbf{Dz} + \mathbf{Bp}/2] > -1$. Finally, we require concavity of $\exp[C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\epsilon})]$, and a sufficient condition for concavity is that $\sum_{l=0}^L z_l \mathbf{A}_l + \mathbf{B}u$ be negative semidefinite. These constraints are assumed to hold for every value that the variables $x, \mathbf{p}, \mathbf{z}, \boldsymbol{\epsilon}$ can take on, and hence every value that $u, \mathbf{p}, \mathbf{z}, \boldsymbol{\epsilon}$ can take on. It is shown in Appendix C that this model can be globally regular if the set $\{x, \mathbf{p}, \mathbf{z}, \boldsymbol{\epsilon}\}$ has bounded support, and that $\boldsymbol{\epsilon}$ can be distributed independently of $x, \mathbf{p}, \mathbf{z}$.

Apart from the construction of y , the implicit Marshallian demand equations (9) are linear in coefficients, which simplifies estimation. In this model, the $\mathbf{D}$ and $\mathbf{B}$ matrix parameters allow for flexible interactions between y and both $\mathbf{z}$ and $\mathbf{p}$. Either or both of these matrices could be zero if such interactions are not needed. Note that if $\mathbf{B}$ were zero, then y in equation (8) would also be linear in parameters.

Our empirical model has Engel curves that are high-order polynomials. Unlike Marshallian demands, our implicit Marshallian EASI demands can be polynomials of any degree without being bound by rank restrictions discussed by Gorman (1981) and Lewbel (1991). Note, however, that monotonicity and concavity of the cost function places inequality constraints on the model, which restricts the range of possible parameter values and the range of values of $\mathbf{p}, y, \mathbf{z}, \boldsymbol{\epsilon}$ for which these demand functions satisfy regularity.

Polynomials are simple but are not required; i.e., we can maintain linearity in coefficients by replacing equation (6) with $\mathbf{m}(y, \mathbf{z}) = \mathbf{Bn}(y, \mathbf{z})$ for some $J \times K$ matrix of constants $\mathbf{B}$ and K -vector of known functions $\mathbf{n}(y, \mathbf{z})$. Our chosen functional form takes $\mathbf{n}(y, \mathbf{z})$ to be a vector of elements of the form u^r and $u^r z_l$, but other functions could also be chosen. For example, the elements of $\mathbf{n}(y, \mathbf{z})$

could be splines or bounded functions such as logistic transformations of polynomials (which would automatically bound estimated budget shares). Semiparametric specifications could be obtained by letting $\mathbf{n}(y, \mathbf{z})$ be basis functions with the number of elements of $\mathbf{n}$ growing to infinity with sample size.

C. Approximate Fully Linear Model

The demand functions (9) are linear in parameters except for the terms $\sum_{l=0}^L z_l \mathbf{p}' \mathbf{A}_l \mathbf{p}$ and $\mathbf{p}' \mathbf{B} \mathbf{p}$ that appear in the construction of y in (8). A similar nonlinearity appears in Deaton and Muellbauer's (1980) AID system and Banks, Blundell, and Lewbel's (1997) QUAID system, and can be dealt with in an analogous way, either by nonlinear estimation or by replacing y with an observable approximation. Consider approximating our real expenditures measure y with nominal expenditures deflated by a Stone price index; that is, replace y with $\tilde{y}$ defined by

$$(10) \quad \tilde{y} = x - \mathbf{p}' \bar{\mathbf{w}}$$

for some set of budget shares $\bar{\mathbf{w}}$. Then, by comparison with equation (9), we have

$$(11) \quad \mathbf{w} = \sum_{r=0}^5 \mathbf{b}_r \tilde{y}^r + \mathbf{C} \mathbf{z} + \mathbf{D} \mathbf{z} \tilde{y} + \sum_{l=0}^L z_l \mathbf{A}_l \mathbf{p} + \mathbf{B} \mathbf{p} \tilde{y} + \tilde{\epsilon},$$

where $\tilde{\epsilon} \approx \epsilon$ with $\tilde{\epsilon}$ defined to make equations (11) hold. We call the model of equations (10) and (11) the Approximate EASI model.

The Approximate EASI nests the model $\mathbf{w} = \mathbf{b}_0 + \mathbf{b}_1 \tilde{y} + \mathbf{C} \mathbf{z} + \mathbf{A} \mathbf{p} + \tilde{\epsilon}$, which is identical to the popular approximate AID if $\tilde{y} = x - \mathbf{p}' \bar{\mathbf{w}}$ (that is, $\bar{\mathbf{w}} = \mathbf{w}$). The AID without the approximation has y equal to deflated x , where the log deflator is quadratic in $\mathbf{p}$, while the EASI model without approximation has y equal to an affine transform of $x - \mathbf{p}' \bar{\mathbf{w}}$.

The approximate EASI also nests the model $\mathbf{w} = \mathbf{b}_0 + \mathbf{b}_1 \tilde{y} + \mathbf{b}_2 \tilde{y}^2 + \mathbf{C} \mathbf{z} + \mathbf{A} \mathbf{p} + \tilde{\epsilon}$, which is the model estimated by Blundell, Panos Pashardes, and Guglielmo Weber (1993). Their motivation for this model was by analogy with the Almost Ideal, but if this model was really to be Marshallian, then, as they show, utility maximization would require linear rank restrictions on the coefficients $\mathbf{b}_0$, $\mathbf{b}_1$, and $\mathbf{b}_2$, which they did not impose. The approximate EASI implicit Marshallian demand function therefore provides a rationale for the unrestricted model that Blundell, Pashardes, and Weber actually estimated.

The approximate EASI model, substituting equation (10) into (11), can be estimated by linear regression methods, with linear cross-equation symmetry restrictions on the $\mathbf{A}_l$ and $\mathbf{B}$ coefficients. A natural choice for $\bar{\mathbf{w}}$ is the sample average of budget shares across consumers. A better approximation to y would be to let $\bar{\mathbf{w}}$ be each consumer's own $\mathbf{w}$, so each consumer has his own Stone index deflator based on his own budget shares. However, as discussed later, this better approximation introduces endogeneity.

In our empirical application, we estimate the approximate model with $\bar{\mathbf{w}} = \mathbf{w}$ using seemingly unrelated regressions, and we estimate the true EASI model using nonlinear 3SLS. As in the approximate AID system, there is no formal theory regarding the quality of the approximation that uses $\tilde{y}$ in place of y , but we find empirically that approximate model estimates do not differ much from estimates based on the exact y (most estimated parameters have the same signs and roughly similar magnitudes), and provide good starting values for exact model estimation.

D. Elasticities and Consumer Surplus

We now show how to evaluate the effects of changing prices or other variables in EASI models. We will give results both for our specific empirical model and for the general EASI model given by equations (2), (3), and (4).

First, consider evaluating the cost to an individual of a price change. A consumer surplus measure for the price change from $\mathbf{p}_0$ to $\mathbf{p}_1$ is the log cost-of-living index, which, for the cost function (2), is given by

$$C(\mathbf{p}_1, u, \mathbf{z}, \boldsymbol{\varepsilon}) - C(\mathbf{p}_0, u, \mathbf{z}, \boldsymbol{\varepsilon}) = (\mathbf{p}_1 - \mathbf{p}_0)' \mathbf{m}(u, \mathbf{z}) + T(\mathbf{p}_1, \mathbf{z}) - T(\mathbf{p}_0, \mathbf{z}) \\ + S(\mathbf{p}_1, \mathbf{z})u - S(\mathbf{p}_0, \mathbf{z})u + (\mathbf{p}_1 - \mathbf{p}_0)' \boldsymbol{\varepsilon}.$$

If $C(\mathbf{p}_0, u, \mathbf{z}, \boldsymbol{\varepsilon})$ is the cost function of a household that has budget shares $\mathbf{w}_0$ and implicit utility level y , then this expression can be rewritten in terms of observables as

$$C(\mathbf{p}_1, u, \mathbf{z}, \boldsymbol{\varepsilon}) - C(\mathbf{p}_0, u, \mathbf{z}, \boldsymbol{\varepsilon}) = (\mathbf{p}_1 - \mathbf{p}_0)' [\mathbf{w}_0 - \nabla_{\mathbf{p}} T(\mathbf{p}_0, \mathbf{z}) - \nabla_{\mathbf{p}} S(\mathbf{p}_0, \mathbf{z})y] \\ + T(\mathbf{p}_1, \mathbf{z}) - T(\mathbf{p}_0, \mathbf{z}) + S(\mathbf{p}_1, \mathbf{z})y - S(\mathbf{p}_0, \mathbf{z})y.$$

For our base empirical model log cost function (5), this log cost-of-living index expression simplifies to

$$C(\mathbf{p}_1, u, \mathbf{z}, \boldsymbol{\varepsilon}) - C(\mathbf{p}_0, u, \mathbf{z}, \boldsymbol{\varepsilon}) = (\mathbf{p}_1 - \mathbf{p}_0)' \mathbf{w}_0 + \frac{1}{2} (\mathbf{p}_1 - \mathbf{p}_0)' \left(\sum_{l=0}^L z_l \mathbf{A}_l + \mathbf{B}y \right) (\mathbf{p}_1 - \mathbf{p}_0).$$

The first term in this cost-of-living index is the Stone index for the price change, $(\mathbf{p}_1 - \mathbf{p}_0)' \mathbf{w}_0$. Such indices are commonly used on the grounds that they are appropriate for small price changes and that they allow for unobserved preference heterogeneity across households. In our model, the presence of the second term, which depends upon T and S , allows us to explicitly model substitution effects, and so consider large price changes, while also accounting for the behavioral importance of both observed and unobserved heterogeneity.

Define semielasticities to be derivatives of budget shares with respect to log prices $\mathbf{p}$, implicit utility y , log nominal total expenditures x , and demographic characteristics (or other observed taste shifters) $\mathbf{z}$. The semielasticity of a budget share can be converted into an ordinary elasticity of a budget share by dividing by that budget share. We provide semielasticities because they are easier to present algebraically. Hicksian demands are given by

$$\boldsymbol{\omega}(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = \mathbf{m}(u, \mathbf{z}) + \nabla_{\mathbf{p}} T(\mathbf{p}, \mathbf{z}) + \nabla_{\mathbf{p}} S(\mathbf{p}, \mathbf{z})u + \boldsymbol{\varepsilon},$$

so the Hicksian price semielasticities are

$$\nabla_{\mathbf{p}} \boldsymbol{\omega}(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = \nabla_{\mathbf{pp}} T(\mathbf{p}, \mathbf{z}) + \nabla_{\mathbf{pp}} S(\mathbf{p}, \mathbf{z})u = \nabla_{\mathbf{pp}} T(\mathbf{p}, \mathbf{z}) + \nabla_{\mathbf{pp}} S(\mathbf{p}, \mathbf{z})y.$$

These are equivalently the price semielasticities holding y fixed, $\nabla_{\mathbf{p}} \boldsymbol{\omega}(\mathbf{p}, y, \mathbf{z}, \boldsymbol{\varepsilon})$. Similarly, semielasticities with respect to y , interpretable as real expenditure elasticities, are given by

$$\nabla_y \boldsymbol{\omega}(\mathbf{p}, y, \mathbf{z}, \boldsymbol{\varepsilon}) = \nabla_y \mathbf{m}(y, \mathbf{z}) + \nabla_{\mathbf{p}} S(\mathbf{p}, \mathbf{z}),$$

and semielasticities with respect to observable demographics $\mathbf{z}$ are

$$\nabla_{\mathbf{z}}\omega(\mathbf{p}, y, \mathbf{z}, \boldsymbol{\varepsilon}) = \nabla_{\mathbf{z}}\mathbf{m}(y, \mathbf{z}) + \nabla_{\mathbf{pz}}T(\mathbf{p}, \mathbf{z}) + \nabla_{\mathbf{pz}}S(\mathbf{p}, \mathbf{z})u.$$

Compensated semielasticities for our baseline model, the log cost function (5), are linear apart from the construction of y . Compensated price semielasticities are given by

$$(12) \quad \nabla_{\mathbf{p}'}\omega(\mathbf{p}, y, \mathbf{z}, \boldsymbol{\varepsilon}) = \sum_{l=0}^L z_l \mathbf{A}_l + \mathbf{B}y,$$

and semielasticities with respect to y are

$$(13) \quad \nabla_y\omega(\mathbf{p}, y, \mathbf{z}, \boldsymbol{\varepsilon}) = \sum_{r=1}^5 \mathbf{b}_r r y^{r-1} + \mathbf{D}\mathbf{z} + \mathbf{B}\mathbf{p},$$

which can vary quite a bit as y changes, reflecting a high degree of Engel curve flexibility. Demographic semielasticities are given by

$$(14) \quad \nabla_{\mathbf{z}'}\omega(\mathbf{p}, y, \mathbf{z}, \boldsymbol{\varepsilon}) = \mathbf{c}_l + \mathbf{d}_l y + \mathbf{A}_l \mathbf{p},$$

where $\mathbf{c}_l$ and $\mathbf{d}_l$ are the appropriate rows of $\mathbf{C}$ and $\mathbf{D}$, respectively, which allows for price and y interactions with demographic effects.

Closed-form expressions for Marshallian elasticities are more complicated, and so are provided in the online Appendix.

II. Estimation

A. Estimators

We estimate demand systems with J goods, so, as usual, we can drop the last equation (the J 'th good) from the system and just estimate the remaining system of $J - 1$ equations. The parameters of the the J 'th good are then recoverable from the adding-up constraint that budget shares sum to one. Assume this is done in all of the following discussion. The system of equations to be estimated is (9).

The approximate EASI, equation (11) with $\tilde{y}$ given by equation (10), is trivial to estimate. If the approximate EASI $\tilde{\boldsymbol{\varepsilon}}$ is uncorrelated with $\mathbf{p}$, $\mathbf{z}$, $\mathbf{z}\tilde{y}$, $\mathbf{p}\tilde{y}$, $\mathbf{p}\mathbf{z}_l$, for $l = 1, \dots, L$ and $\tilde{y}^r$ for $r = 0, \dots, 5$, then, without imposing symmetry of the $\mathbf{A}_l$ and $\mathbf{B}$ matrices, estimating each approximate EASI equation separately by linear ordinary least squares is consistent and equivalent to linear seemingly unrelated regressions (SUR). Imposing symmetry of the $\mathbf{A}_l$ and $\mathbf{B}$ matrices means imposing linear cross-equation equality constraints on the coefficients, but the resulting approximate EASI model can still be consistently estimated using ordinary linear SUR. Since this model is only an approximation to the EASI model, we should not expect these uncorrelatedness assumptions to hold exactly, but we found that the approximate EASI estimates were generally quite close to the exact model estimates, and can provide useful parameter starting values for exact model estimation.

The exact EASI model (without approximation) that we estimate has equation (8) substituted into equation (9) to give

$$(15) \quad \mathbf{w} = \sum_{r=0}^5 \mathbf{b}_r \left(\frac{x - \mathbf{p}'\mathbf{w} + \sum_{l=0}^L z_l \mathbf{p}'\mathbf{A}_l \mathbf{p}/2}{1 - \mathbf{p}'\mathbf{B}\mathbf{p}/2} \right)^r + \mathbf{C}\mathbf{z} + \sum_{l=0}^L z_l \mathbf{A}_l \mathbf{p} \\ + (\mathbf{D}\mathbf{z} + \mathbf{B}\mathbf{p}) \left(\frac{x - \mathbf{p}'\mathbf{w} + \sum_{l=0}^L z_l \mathbf{p}'\mathbf{A}_l \mathbf{p}/2}{1 - \mathbf{p}'\mathbf{B}\mathbf{p}/2} \right) + \boldsymbol{\varepsilon}.$$

Equation (15) is nonlinear in parameters because of the presence of $\mathbf{A}_l$ and $\mathbf{B}$ in y . An additional complication for estimation is that $\mathbf{w}$, which is endogenous, appears on the right side of equation (15) in the Stone index $\mathbf{p}'\mathbf{w}$. One possible estimator is to ignore this endogeneity and simply estimate the resulting system of equations by nonlinear least squares. In our empirical application we find that the resulting bias from ignoring this endogeneity is small, because, as we discuss below, the variation in the $\mathbf{p}'\boldsymbol{\varepsilon}$ component of $\mathbf{p}'\mathbf{w}$ is tiny relative to the variation in x and the other components of y .

To formally account for endogeneity, nonlinearity in parameters, and possible heteroskedasticity of unknown form in $\boldsymbol{\varepsilon}$, we use an instrumental variables estimator. Let $\mathbf{q}$ be an M -vector of observable variables that are uncorrelated with $\boldsymbol{\varepsilon}$, which will be used as instruments for estimation. If $E(\boldsymbol{\varepsilon} | \mathbf{x}, \mathbf{p}, \mathbf{z}) = \mathbf{0}_J$, then $\mathbf{q}$ can include any bounded functions of $\mathbf{p}$, $\mathbf{z}$, and x , for example, $\mathbf{p}$, $\mathbf{z}$, and $\tilde{y} = x - \mathbf{p}'\bar{\mathbf{w}}$ with $\bar{\mathbf{w}}$ equal to the sample average budget-share vector. However, if unobserved heterogeneity is correlated with some observed characteristics such as x or elements of $\mathbf{z}$, then those elements must be excluded from $\mathbf{q}$. Let $\mathbf{q} = (q_1, \dots, q_M)'$. Then $E[\boldsymbol{\varepsilon}'q_m] = \mathbf{0}_J$ for $m = 1, \dots, M$, which implies

$$(16) \quad E \left[\begin{pmatrix} \mathbf{w} - \sum_{r=0}^5 \mathbf{b}_r \left(\frac{x - \mathbf{p}'\mathbf{w} + \sum_{l=0}^L z_l \mathbf{p}'\mathbf{A}_l \mathbf{p}/2}{1 - \mathbf{p}'\mathbf{B}\mathbf{p}/2} \right)^r - \mathbf{C}\mathbf{z} - \sum_{l=0}^L z_l \mathbf{A}_l \mathbf{p} \\ - (\mathbf{D}\mathbf{z} + \mathbf{B}\mathbf{p}) \left(\frac{x - \mathbf{p}'\mathbf{w} + \sum_{l=0}^L z_l \mathbf{p}'\mathbf{A}_l \mathbf{p}/2}{1 - \mathbf{p}'\mathbf{B}\mathbf{p}/2} \right) \end{pmatrix} q_m \right] = \mathbf{0}_J$$

for $m = 1, \dots, M$.

Allowing for generally heteroskedastic $\boldsymbol{\varepsilon}$, parameters may be estimated by applying Hansen's (1982) GMM to this set of moment conditions. By adding up, the moments associated with the J 'th good are superfluous, so with M instruments we will have $(J - 1)M$ moments. An alternative, simpler estimator (which we found to be numerically better behaved in practice) is homoskedastic nonlinear 3SLS which is, like GMM or heteroskedastic 3SLS, consistent with heteroskedastic $\boldsymbol{\varepsilon}$, but is only as asymptotically efficient as these other estimators when $\boldsymbol{\varepsilon}$ is homoskedastic.

Another possible estimator (suggested by a referee) is a form of iterative linear 3SLS, as follows. Let $\boldsymbol{\theta}$ be the vector of all the parameters in the model. If y were observed, we could estimate $\boldsymbol{\theta}$ by a linear three stage least squares, that is, we could estimate the system of linear equations (9) using the vector $\mathbf{q}$ as instruments. Given observations (consumers) $y_1, \dots, y_n$, let $\hat{\boldsymbol{\theta}} = \boldsymbol{\Theta}_n(y_1, \dots, y_n)$ denote this linear three stage least squares estimator, or any other linear model GMM estimator based on the moments $E[\boldsymbol{\varepsilon}'q_m] = \mathbf{0}_J$ for $m = 1, \dots, M$ with $\boldsymbol{\varepsilon}$ given by equation (9). For each consumer i , we have $\hat{y}_i = g(\mathbf{w}_i, \mathbf{p}_i, x_i, \mathbf{z}_i, \hat{\boldsymbol{\theta}})$ for the function g defined by equation (8). The estimator would then consist of taking some starting values for $\hat{\boldsymbol{\theta}}$ (these could be estimates using our approximate model estimator of $\boldsymbol{\theta}$), substituting this $\hat{\boldsymbol{\theta}}$ into g to obtain estimates $\hat{y}_1, \dots, \hat{y}_n$, do linear three stage least squares or linear model GMM using these $\hat{y}$ s as data to obtain a new

value of θ given by $\hat{\theta} = \Theta_n(\hat{y}_1, \dots, \hat{y}_n)$, and repeat this process using $\hat{\theta}$ in place of θ , iterating to convergence. This estimator could be easily implemented in a statistical package like Stata that does linear three stage least squares or linear GMM regression. Note that imposing symmetry restrictions requires imposing cross-equation equality restrictions on the estimated coefficients in each iteration.

Formally, let the fixed point $\hat{\theta} = \Theta_n[(g(\mathbf{w}_1, \mathbf{p}_1, x_1, \mathbf{z}_1, \hat{\theta}), \dots, g(\mathbf{w}_n, \mathbf{p}_n, x_n, \mathbf{z}_n, \hat{\theta}))]$ define the estimator $\hat{\theta}$. This is a special case of the generic class of fixed point based estimators considered by Jeff Dominitz and Robert Sherman (2005), who provide associated consistency and limiting distribution theory assuming that the mapping that defines the estimator $\hat{\theta}$ is a contraction mapping. If the iterations had been linear least squares seemingly unrelated regressions instead of linear three stage least squares regressions, then this estimator would also be an example of the iterated linear least squares estimator for conditionally linear systems proposed by Blundell and Jean-Marc Robin (1999). Blundell and Robin apply the estimator to a demand system context similar to ours and provide an extension to endogenous regressors, but their extension requires a control function form of endogeneity that our model does not satisfy. We do not provide formal asymptotic theory for $\hat{\theta}$ in our model, but we found that this estimator performed very well in practice, yielding estimates that are numerically quite close to those of our full nonlinear three stage least squares estimator.

The equality restrictions required for demand system rationality are easily imposed in our context. Adding up and homogeneity constraints on the parameter vectors and matrices hold by omitting the J 'th good and imposing the linear restrictions that $\mathbf{1}'\mathbf{A}_l = \mathbf{1}'\mathbf{B} = \mathbf{0}'$. Slutsky symmetry is satisfied if and only if $\mathbf{A}_l$ for $l = 0, \dots, L$ and $\mathbf{B}$ are all symmetric matrices. All of these parametric restrictions may be imposed as a set of linear constraints in 3SLS or GMM.

Writing this system linearly as equation (9) suggests that good instruments $\mathbf{q}$ should be highly correlated with $\mathbf{p}$, $\mathbf{z}$, $\mathbf{zy}$, $\mathbf{py}$, $\mathbf{pz}_1, \dots, \mathbf{pz}_L$, and $\mathbf{y}'$. We assume $E(\boldsymbol{\varepsilon} | \mathbf{p}, \mathbf{z}) = \mathbf{0}_J$ and take $\mathbf{q}$ to be $\mathbf{p}$, $\mathbf{z}$, $\mathbf{zy}$, $\mathbf{py}$, $\mathbf{pz}_1, \dots, \mathbf{pz}_L$, and $\bar{\mathbf{y}}'$ for $r = 0, \dots, 5$, with $\bar{\mathbf{y}}$ defined as

$$\bar{\mathbf{y}} = \frac{x - \mathbf{p}'\bar{\mathbf{w}} + \sum_{l=0}^L z_l \mathbf{p}'\bar{\mathbf{A}}_l \mathbf{p} / 2}{1 - \mathbf{p}'\bar{\mathbf{B}}\mathbf{p} / 2},$$

where $\bar{\mathbf{w}}$ is the average budget shares across consumers in our sample, and $\bar{\mathbf{A}}_l$ and $\bar{\mathbf{B}}$ are the estimated values of $\mathbf{A}_l$ and $\mathbf{B}$ based on linear least squares estimation of the approximate EASI model. Note that use of $\bar{\mathbf{w}}$ and inconsistency of the estimates of $\bar{\mathbf{A}}_l$ and $\bar{\mathbf{B}}$ (due to their coming from the approximate model) in the construction of $\bar{\mathbf{y}}$ affects only the quality of the instruments $\mathbf{q}$ and hence the efficiency of the 3SLS or GMM estimation, but does not cause inconsistency, because $\bar{\mathbf{y}}$ remains uncorrelated with $\boldsymbol{\varepsilon}$ given any choice of values of the parameters $\bar{\mathbf{w}}$, $\bar{\mathbf{A}}_l$, and $\bar{\mathbf{B}}$ in the construction of $\bar{\mathbf{y}}$. When symmetry of $\mathbf{A}_l$ and $\mathbf{B}$ is not imposed, this set of moments $E(\boldsymbol{\varepsilon}'\mathbf{q}_m) = \mathbf{0}_J$ for $m = 1, \dots, M$ exactly identifies the EASI model parameters. Imposing symmetry reduces the number of distinct parameters, yielding overidentification.

Since we have assumed $E(\boldsymbol{\varepsilon} | \mathbf{p}, \mathbf{z}) = \mathbf{0}_J$, we could interpret $\sum_{r=0}^5 \mathbf{b}_r \mathbf{y}^r + \mathbf{Cz} + \mathbf{Dzy}$ as a sieve approximation to a general unknown J -vector of smooth functions $\mathbf{n}(\mathbf{y}, \mathbf{z})$, with a spanning basis consisting of functions of the form $\mathbf{y}^s \mathbf{z}_k^t$ for integers s, t . Chunrong Ai and Xiaohong Chen (2003) provide associated rates of convergence, optimal instrument construction, and limiting distribution theory for general semiparametric sieve GMM estimators of this form. Such estimators can attain the semiparametric efficiency bound for the remaining parameters, in this case $\mathbf{A}_l$ and $\mathbf{B}$.

The parametric 3SLS or GMM estimators can be readily modified to deal with possible measurement error or endogeneity in some regressors, by suitably modifying the set of instruments $\mathbf{q}$. For example, if simultaneity with supply is a concern (which is more likely to matter

significantly in an aggregate demand context than in our empirical application), then we could replace $\mathbf{p}$ with $\bar{\mathbf{p}}$ everywhere $\mathbf{p}$ appears in the construction of the instruments $\mathbf{q}$, where $\bar{\mathbf{p}}$ are the fits from regressing $\mathbf{p}$ on supply-side instruments.

In many datasets, such as the UK Family Expenditure Survey, consumption is measured over a very short time span and so is subject to considerable infrequency-based measurement error. In our empirical application, we use Canadian data where consumption is measured over an entire year, which implies less infrequency-based error, but may suffer from recall-based error. One might deal with this problem in part by including functions of income in place of functions of nominal total expenditures in our list of instruments. However, using an instrument like income, which has a large amount of over-time variation that is smoothed out in consumption decisions, may entail considerable efficiency loss. One might alternatively use measures of household wealth as instruments for total expenditure but, unfortunately, most public-use datasets (including ours) do not have both wealth and consumption data. It should also be noted that measurement errors in total expenditures would affect not only our total expenditures regressor, but also the construction of our dependent variables, the budget shares. Estimators such as Lewbel (1996) could potentially be used to deal with the latter problem.

Similarly, if unobserved preference heterogeneity $\boldsymbol{\varepsilon}$ is correlated with some observed taste shifters (i.e., elements of $\mathbf{z}$), then those may be excluded from the instrument list and replaced by, e.g., nonlinear functions of x and of the remaining elements of $\mathbf{z}$. However, in these cases one would need to take care in interpreting the resulting model residuals $\hat{\boldsymbol{\varepsilon}}$, because they will then contain both unobserved preference heterogeneity and measurement error. With panel data one might separate these two effects by modeling the unobserved preference heterogeneity using standard random or fixed effects methods. The exact model estimators remain consistent regardless of heteroskedasticity in $\boldsymbol{\varepsilon}$, so, for example, the estimates are consistent if $\boldsymbol{\varepsilon} = \mathbf{N}(x, \mathbf{z})\boldsymbol{\varepsilon}^*$ where $\boldsymbol{\varepsilon}^*$ are preference parameters that are independent of $\mathbf{p}, x, \mathbf{z}$, and features of the function $\mathbf{N}(x, \mathbf{z})$ could be estimated based on the estimated conditional variance of residuals $\hat{\boldsymbol{\varepsilon}}$, conditioning on $x, \mathbf{z}$.

The estimators described above do not impose the inequality (concavity and monotonicity) constraints implied by regularity of the cost function, or equivalently by utility maximization (global regularity is discussed in Appendix C). In addition to these usual utility-derived inequality constraints, interpreting $\boldsymbol{\varepsilon}$ as preference heterogeneity parameters requires $\boldsymbol{\varepsilon}$ to be independent of $\mathbf{p}$, and having the support of $\boldsymbol{\varepsilon}$ independent of prices imposes additional inequality constraints (see the online Appendix for details). If the model is correctly specified, then imposing these inequality constraints on parameter estimates will not be binding asymptotically, and so failing to impose them will not result in a loss of asymptotic efficiency, as long as the true parameter values do not lie on the boundary of the feasible region implied by the inequality constraints. Based on this, a common practice in empirical demand analysis, which we will follow in our application, is to estimate the model without imposing inequality constraints, and then check that they are satisfied for a reasonable range of $\mathbf{p}, x$, and $\mathbf{z}$ values. In particular, the inequalities associated with utility function regularity are readily verified using elasticity calculations, e.g., the estimated Slutsky matrix can be checked for negative semidefiniteness.

B. Data and Model Tests

We first estimate the approximate EASI model, equation (11) with $\tilde{y}$ given by equation (10), using SUR. We then estimate the exact EASI model given by (15) by nonlinear 3SLS with instruments and starting values as described in the previous section. We use 3SLS because we found the GMM weighting matrix estimates to be numerically unstable. Also, 3SLS is asymptotically efficient under the assumption of independently normally distributed $\boldsymbol{\varepsilon}$, and our 3SLS estimator remains consistent (though inefficient) if $\boldsymbol{\varepsilon}$ is not normal or homoskedastic.

TABLE 1—DATA DESCRIPTIVES

Variable		Mean	Std Dev	Minimum	Maximum
Budget shares	Food at home	0.14	0.08	0.00	0.61
	Food out	0.08	0.07	0.00	0.63
	Rent	0.37	0.12	0.01	0.89
	Household operation	0.07	0.04	0.00	0.61
	Household furnishing	0.04	0.05	0.00	0.52
	Clothing	0.08	0.06	0.00	0.48
	Transportation operation	0.11	0.08	0.00	0.60
	Recreation	0.08	0.07	0.00	0.59
Log-prices	Personal care	0.03	0.02	0.00	0.21
	Food at home	−0.05	0.43	−1.41	0.34
	Food out	0.04	0.49	−1.46	0.53
	Rent	−0.08	0.40	−1.27	0.37
	Household operation	−0.06	0.45	−1.40	0.32
	Household furnishing	−0.05	0.32	−0.94	0.20
	Clothing	0.04	0.35	−0.94	0.34
	Transportation operation	−0.07	0.57	−1.53	0.57
Demographics	Recreation	0.01	0.40	−1.04	0.42
	Personal care	−0.03	0.38	−1.11	0.29
	Age minus 40	0.71	11.89	−15.00	24.00
	Male	0.51	0.50	0.00	1.00
	Car owner	0.42	0.49	0.00	1.00
Log-expenditure (median-norm'd)	Social assistance	0.27	0.44	0.00	1.00
	Time	88.99	8.73	69.00	99.00
Log-expenditure (median-norm'd)	x	−0.11	0.59	−2.75	1.66
	$x - \mathbf{p}'\mathbf{w}$	−0.07	0.44	−1.70	1.44

The data used in this paper come from the following public use sources: (1) the Family Expenditure Surveys 1969, 1974, 1978, 1982, 1984, 1986, 1990, 1992, and 1996; (2) the Surveys of Household Spending 1997, 1998, and 1999; and (3) Browning and Irene Thomas (1999), with updates and extensions to rental prices from Pendakur (2002). Price and expenditure data are used from 12 years in 4 Canadian regions (Atlantic, Ontario, Prairies, and British Columbia), yielding 48 distinct price vectors. Prices are normalized so that the price vector facing residents of Ontario in 1986 is $(1, \dots, 1)$, that is, these observations define the base price vector $\bar{\mathbf{p}} = \mathbf{0}_J$. Since the model contains a high-order polynomial in y , we subtract the median value of $x - \mathbf{p}'\mathbf{w}$ from y . Translating y by a constant is absorbed by changes in the $\mathbf{A}_0$, $\mathbf{b}_n$, and $\mathbf{C}$ coefficients leaving the fit unchanged, and shifting y to near zero reduces numerical problems associated with data matrix conditioning by giving the $\mathbf{b}_r$ coefficients comparable magnitudes.

The empirical analysis uses annual expenditure in $J = 9$ expenditure categories: food-in, food-out, rent, clothing, household operation, household furnishing and equipment, transportation operation, recreation, and personal care. Personal care is the left-out equation, yielding eight expenditure share equations to be estimated. These expenditure categories, which exclude large durables, account for about 85 percent of the current consumption of the households in the sample.

Our sample for estimation consists of 4,847 observations of rental-tenure single-member households that had positive expenditures on rent, recreation, and transportation. We use only persons age 25 to 64 living in cities with at least 30,000 residents. We use residents of English-speaking Canada only (so Quebec is excluded). Single-member households are used to avoid issues associated with bargaining or other complications that may be associated with the behavior of collective households. City residents are used to minimize the effects of possible home and farm production. Table 1 gives summary statistics for our estimation sample.

We include $L = 5$ observable demographic characteristics in our model: (1) the person's age minus 40; (2) the sex dummy equal to one for men; (3) a car-nonowner dummy equal to one if real gasoline expenditures (at 1986 gasoline prices) are less than \$50; (4) a social assistance dummy equal to one if government transfers are greater than 10 percent of gross income; and (5) a time variable equal to the calendar year minus 1986 (that is, equal to zero in 1986).

We include the time variable in the model to account for possible slow changes over time in tastes, quality, and composition of composite goods. Time trends are also commonly included in demand systems to account for some types of habit formation (see, e.g., Robert Pollak and Terence Wales 1992). The time trend may also help to compensate for average growth in, and hence possible nonstationarity of, real total expenditures over time, noting that our dependent variables, budget shares, are bounded. See, e.g., Lewbel and Serena Ng (2005) for a detailed analysis of the effects of covariate nonstationarity in budget-share demand-system models. However, our data consist of repeated cross sections at the household level, so the scope for possible dependence of errors across observations due to total expenditure growth is limited.

These variables are all zero for a 40-year-old car-owning female in 1986 who did not receive much government transfer income. We use a limited set of characteristics like this because each additional element of $\mathbf{z}$ increases the number of additional parameters to estimate in our model by 52 (80 additional parameters minus 28 additional symmetry restrictions). One could introduce additional demographics at a lower cost in degrees of freedom by including a large set of characteristics in the $\mathbf{m}$ function (the $\mathbf{C}$ and $\mathbf{D}$ parameter matrices) and a smaller set interacted with prices (the $\mathbf{A}_l$ matrices). Restrictiveness of the observed heterogeneity specification is also partly mitigated by the relative demographic homogeneity of the sample, and by the role of $\boldsymbol{\varepsilon}$, which embodies unobserved preference heterogeneity.

When symmetry is not imposed on $\mathbf{A}_l$ and $\mathbf{B}$, then, given the instruments specified above, the model is exactly identified. Denoting the number of terms in each $\mathbf{b}_r$ (six in our baseline model) as R , the symmetry-unrestricted model has $[R + L + L + (J - 1)(L + 1) + (J - 1)](J - 1) = 576$ parameters for $\mathbf{b}_r$, $\mathbf{C}$, $\mathbf{D}$, $\mathbf{A}_l$, and $\mathbf{B}$ (72 coefficients in each of 8 equations). When symmetry is imposed on $\mathbf{A}_l$ and $\mathbf{B}$, the model has $(L + 2)(J - 1)(J - 2)/2 = 196$ restrictions, which are overidentifying restrictions for testing exogeneity and are also symmetry restrictions. Since each estimated model has hundreds of parameters, we do not present all the individual coefficients estimates (they are available from the authors on request). We instead provide many summaries and analyses of the results. Even with parameters all set to zero as starting values, we still obtained convergence of the nonlinear 3SLS exact model estimator in only three iterations, due to the near linearity of the EASI model.

Table 2 gives Wald- and J-tests for various hypotheses concerning the model. Since our sample size is large (8 equations times 4,847 observations per equation), we use a 1 percent critical value for all tests. Consider price effects first. The Wald test of symmetry in the asymmetric model is marginally insignificant with a p -value of 1.4 percent, and symmetry is not rejected for either the level of prices ($\mathbf{A}_l = \mathbf{A}_l'$ for all l) or for prices interacted with implicit utility y ($\mathbf{B} = \mathbf{B}'$). This lack of symmetry rejection is not due to the irrelevance of price effects. In particular, the direct price effects given by the $\mathbf{A}_0$ matrix are strongly significant, as are many of the interactions between prices and demographics in the $\mathbf{A}_l$ matrices. The interaction of prices with income are less significant, and we cannot reject the hypothesis that the $\mathbf{py}$ interaction given by the $\mathbf{B}$ matrix is excludable.

Turning to expenditure effects, we first check for adequacy of our fifth-order polynomial in y by adding either a y^{-1} or y^6 term to the model. The reciprocal of y is excludable with a p -value of 4.9 percent. The sixth-order term in y is marginally significant, but these tests covary, and the joint test of exclusion of both y^{-1} and y^6 is not significant, with a p -value of 2.9 percent. Taken together, the results above suggest that symmetry may not be violated and that a fifth-order

TABLE 2—MODEL TESTS (WALD- OR J-TESTS)

Model	Test of	Parameters	df	Test Stat	p-value
Asymmetric with y^{-1}, y^6	symmetry	$A_l = A_l'$ for all l ; $B = B'$	196	241.8	0.014
	symmetry	$A_l = A_l'$ for all l	168	194.1	0.082
	symmetry	$B = B'$	28	24.4	0.661
	exclusion	$B = 0$	64	82.5	0.060
	exclusion	$A_l = 0$ for all l	384	636.0	0.000
	exclusion	$B = 0, A_l = 0$ for all l	448	999.7	0.000
	exclusion	y^{-1}	8	15.6	0.049
	exclusion	y^6	8	20.6	0.008
	exclusion	y^5	8	21.0	0.007
	exclusion	y^{-1}, y^6	16	28.3	0.029
	exclusion	y^{-1}, y^6, y^5	24	75.5	0.000
Symmetric with without y^{-1}, y^6	exclusion	$B = 0$	36	51.7	0.043
	exclusion	$A_l = 0$ for all l	216	646.2	0.000
	exclusion	$B = 0, A_l = 0$ for all l	252	710.6	0.000
	exclusion	y^5	8	45.8	0.000
	non-quadratic	Food at home	3	16.9	0.001
	non-quadratic	Food out	3	10.3	0.016
	non-quadratic	Rent	3	94.4	0.000
	non-quadratic	Household operations	3	22.7	0.000
	non-quadratic	Household furnishing	3	3.4	0.340
	non-quadratic	Clothing	3	9.4	0.025
	non-quadratic	Transportation operation	3	6.8	0.079
	non-quadratic	Recreation	3	28.9	0.000
	overidentification	J-test (with sym pz)	196	238.4	0.021
	overidentification	J-test (with sym. pz, y^{-1}, y^6)	212	273.4	0.003

polynomial in y may be sufficient, so we present further results for a symmetry-restricted model with $r = 0, 1, \dots, 5$.

The lower panel of Table 2 shows that one cannot further reduce the level of the polynomial in y by dropping y^5 . Turning to evidence of complicated Engel curve shapes, we test for whether each of the eight budget share equations can be reduced to a quadratic. The tests suggest that four of the eight budget shares are statistically significantly nonquadratic: food at home, rent, household operation, and recreation. These departures from quadratic Engel curves suggest that allowing for complex Engel curves is a useful attribute of our methodology. Later we examine our estimated Engel curve shapes in more detail, considering whether the departures from quadratic are economically important in addition to being statistically significant.

Next, consider some model summary statistics. The symmetry-restricted model is overidentified, so we can construct overidentification tests of instrument validity. The 196 symmetry-restrictions result in our model having 196 more moments than parameters, and we pass the resulting overidentification-based test of validity of the included moments with a p -value of 2.1 percent. However, if we include additional instruments based on y^{-1} and y^6 , we obtain 16 additional moments (two more instruments times eight budget share equations), and the resulting J-test of overidentifying restrictions becomes marginally significant.

Let e denote the sample residuals vector, which is also our estimate of ϵ for each household. The R^2 values for the eight equations (in the order given in Table 2) are: 0.49, 0.31, 0.65, 0.30, 0.19, 0.43, 0.57, and 0.31, respectively. The model is not estimated by ordinary least squares, so these R^2 s should be interpreted only as summary measures regarding the size of e relative to budget shares. These numbers show that much, even most, of the variation in budget shares is due to unobserved heterogeneity. Our approach formally includes ϵ as random utility parameters, so the relatively large variation in e yields large effects in our welfare calculation estimates, as we demonstrate below.

TABLE 3—VARIATION AND CORRELATION OF y COMPONENTS
(Detrended variables)

Variable	std dev	y	$y(\mathbf{p}'(\mathbf{w}-\mathbf{e}))$	$y\text{-bar}$	$x-\mathbf{p}'\mathbf{w}$	$x-\mathbf{p}'\mathbf{w}\text{-bar}$	$\mathbf{p}'\mathbf{e}$	$\mathbf{p}'(\mathbf{w}-\mathbf{e})$	$\mathbf{p}'\mathbf{A}\mathbf{p}/2$	$\mathbf{p}'\mathbf{B}\mathbf{p}/2$
y	0.4393	1.0000								
$y(\mathbf{p}'(\mathbf{w}-\mathbf{e}))$	0.4390	0.9988	1.0000							
$y\text{-bar}$	0.4459	0.9984	0.9996	1.0000						
$x-\mathbf{p}'\mathbf{w}$	0.4387	1.0000	0.9988	0.9984	1.0000					
$x-\mathbf{p}'\mathbf{w}\text{-bar}$	0.4452	0.9984	0.9996	1.0000	0.9984	1.0000				
$\mathbf{p}'\mathbf{e}$	0.0213	-0.0445	0.0041	0.0035	-0.0444	0.0035	1.0000			
$\mathbf{p}'(\mathbf{w}-\mathbf{e})$	0.0716	0.0678	0.0672	0.0741	0.0685	0.0748	-0.0141	1.0000		
$\mathbf{p}'\mathbf{A}\mathbf{p}/2$	0.0018	0.0173	0.0165	0.0234	0.0134	0.0196	-0.0172	-0.2243	1.0000	
$\mathbf{p}'\mathbf{B}\mathbf{p}/2$	0.0011	-0.0123	-0.0141	-0.0122	-0.0140	-0.0140	-0.0371	-0.6439	0.4676	1.0000

Table 3 provides information on the sources of variation in y . Since both total expenditures and prices may contain time trends, we detrended all the variables in this table by regressing them on dummies for each time period, and report correlations of the resulting residuals from these regressions. The upper block of Table 3 gives standard deviations and pairwise correlations between three variables. The first is y itself, detrended. The second, denoted $y(\mathbf{p}'(\mathbf{w} - \mathbf{e}))$ is identical to y except that it removes $\mathbf{p}'\mathbf{e}$ from the $\mathbf{p}'\mathbf{w}$ component of y given in equation (8). The third is the instrument $\tilde{y}$, which is identical to y , except that it has $\mathbf{p}'\bar{\mathbf{w}}$, where $\bar{\mathbf{w}}$ is the sample-average budget-share vector in the numerator instead of $\mathbf{p}'\mathbf{w}$. These three variables have almost identical variances and correlations over 0.998, even after detrending. This shows that the instrument $\tilde{y}$ is highly relevant and that, while our estimator controls for endogeneity in y , it may not be numerically important to do so in practice, since removing the source of endogeneity from y leaves it virtually unchanged.

Further examining the components of y , the second block of Table 3 shows correlations between y and two measures of the log of stone-deflated expenditures, $x - \mathbf{p}'\mathbf{w}$ and $\tilde{y} = x - \mathbf{p}'\bar{\mathbf{w}}$. Even after detrending, these variables are also highly correlated with each other, which explains why the approximate EASI model estimates are close to the exact model estimates, even though the approximate EASI does not include the affine transformation of $x - \mathbf{p}'\mathbf{w}$ that defines $\tilde{y}$, and our approximate EASI estimates do not control for endogeneity.

The third block of Table 3 considers the magnitude of $\mathbf{p}'\mathbf{e}$ in more detail. Here, we see that the standard deviation of $\mathbf{p}'\mathbf{e}$ is about one-twentieth as large as that of y and of the log of stone-deflated expenditures, $x - \mathbf{p}'\mathbf{w}$, and that $\mathbf{p}'\mathbf{e}$ is only weakly correlated with these real expenditure measures. This again shows that the endogenous component of y or $\tilde{y}$ is small. It also shows that if the errors in our model were treated as measurement errors in budget shares (or, more generally, as errors that are nonbehavioral or otherwise not part of the model), then our parameter estimates (including our finding of significantly nonquadratic Engel curves) would be almost unchanged. However, treating the errors as measurement errors rather than preference heterogeneity parameters substantially affects consumer surplus and social welfare calculations, as we show later in the section on consumer surplus estimates. Formally excluding $\mathbf{p}'\mathbf{e}$ from y requires iterative estimation, following the spirit of the iterative nonparametric approach proposed in Pendakur and Sperlich (2008). We discuss the results of such estimation at the end of Section II.

The last block of Table 3 gives the standard deviation and correlations of $\sum_{i=0}^L z_i \mathbf{p}'\mathbf{A}_i \mathbf{p}/2$ (denoted $\mathbf{p}'\mathbf{A}\mathbf{p}$ in the Table) and $\mathbf{p}'\mathbf{B}\mathbf{p}/2$, which are the components of the difference between y and $x - \mathbf{p}'\mathbf{w}$. These components have less than one-tenth the standard deviation of $\mathbf{p}'\mathbf{e}$ and less than a hundredth the standard deviation of y . As a result, they drive only a tiny fraction of the variation in y , which again shows why the approximation that ignores these components is quite good.

C. Estimated Engel Curves

The easiest way to summarize the income-related parameter estimates is to examine the resulting expenditure share equations as functions of x for particular values of $\mathbf{p}$, $\mathbf{z}$, and $\boldsymbol{\varepsilon}$. At $\mathbf{p} = \mathbf{0}_j$, y equals log nominal expenditure x , so at these base prices we obtain Marshallian Engel curves $\mathbf{w} = \sum_{r=0}^5 \mathbf{b}_r x^r + \mathbf{Cz} + \mathbf{Dz}x + \boldsymbol{\varepsilon}$. Figures 1–8 show these estimated Engel curves from our model for a 40-year-old car-owning female in 1986 who did not receive much government transfer income, and having $\boldsymbol{\varepsilon} = \mathbf{0}$. For her, $\mathbf{w} = \sum_{r=0}^5 \mathbf{b}_r x^r$. The base-period Engel curves for households with different values of unobserved heterogeneity are identical except for being vertically shifted by $\boldsymbol{\varepsilon}$. These base-period Engel curves are also informative about the shape of Engel curves in other price regimes, since at other price vectors, Engel curves expressed in terms of our real expenditures measure y differ from the above only by the addition of the linear function $\sum_{l=0}^L \mathbf{A}_l z_l \mathbf{p} + \mathbf{Bp}y$.

In Figures 1–8, each figure presents the estimated Engel curve for four models: thick black lines indicate 3SLS estimates of the symmetry-restricted *exact* model; thick gray lines indicate 3SLS estimates of the symmetry-restricted *quadratic* (having a second- instead of fifth-order polynomial in y) exact model; thin black lines indicate *approximate* SUR estimates of the model; and medium black lines indicate exact 3SLS estimates of the *asymmetric* model (*italics* denote shorthand). Estimates are computed at each percentile of log expenditures in the data, and estimated 90 percent confidence intervals (computed via the delta method) are displayed with small crosses for the exact 3SLS symmetry-restricted model at each fifth percentile of the expenditure distribution.

As noted earlier, our quadratic specification is very similar to Quadratic Almost Ideal (QAI) related models estimated by Blundell, Pashardes, and Weber (1993) and Banks, Blundell, and Lewbel (1997). The quadratic EASI differs from these QAI models in that it has a different deflator for total expenditures, and it does not require the coefficient of squared y to depend on either prices or on the lower-order expenditure coefficients to stay consistent with utility maximization. Our specification also allows for more general demographic effects than is typical in applied QAI models, which usually have terms analogous to $\mathbf{C}$, but not to $\mathbf{A}_l$ or $\mathbf{D}$.

Figures 1 and 2 show Engel curves for food-in (food consumed at home) and food-out. Both these Engel curves are almost linear. Both the approximate and quadratic symmetry-restricted models lie within the 90 percent pointwise confidence intervals of the exact model. However, in the food-out equation, the asymmetric model estimates are above the confidence intervals in the upper part of the expenditure distribution. The food-in budget share equation is statistically significantly nonquadratic (see Table 2), though the empirical size of the departure is small.

Figure 3 shows the Engel curve for rent. Throughout most of the expenditure distribution, the quadratic estimates lie outside the confidence intervals of the exact estimates. In the exact estimates, the curvature of the Engel curve is near zero in the bottom decile, strongly negative in the next two deciles, and near zero until it becomes positive in the top decile. This kind of complexity cannot be captured in a quadratic model but is readily accommodated in the EASI framework. Further, the magnitude of the quadratic departure from the exact model can be large. At the fifth percentile, the quadratic model overestimates the rent budget share by almost 5 percentage points. At the bottom quintile cutoff, the quadratic model underestimates the rent budget share by about 2 percentage points.

For the rent equation, the approximate model performs very well, lying essentially on top of the exact estimates throughout the distribution of expenditures. As in the food-out equation, the asymmetric model performs relatively poorly (though not as poorly as the quadratic model).

Figures 4 through 7 give the household operation, household furnishing and equipment, clothing, and transportation operation Engel curves. All four sets of estimates lie within the 90

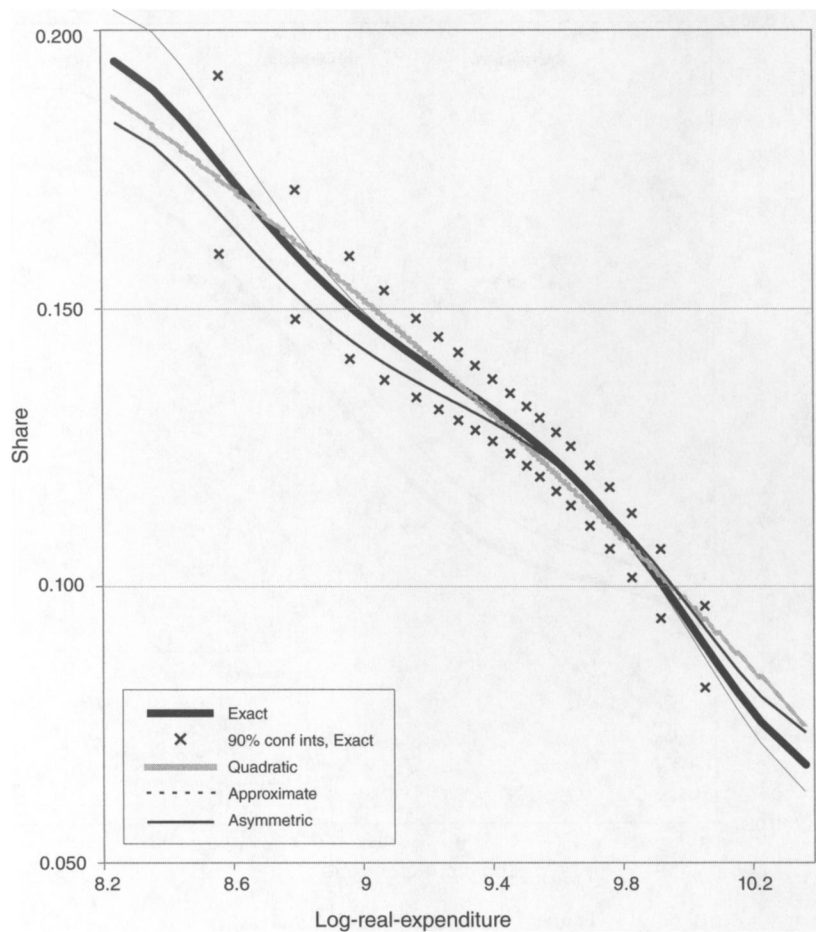


FIGURE 1. ESTIMATED FOOD-IN SHARES

percent confidence intervals of the exact model estimates. However, as shown in Table 2, the household operation equation shows evidence of being statistically significantly nonquadratic. The departure from quadratic can be seen in Figure 4, where the exact model estimate looks more like two nearly linear segments joined by a curved segment in the middle instead of a true quadratic function.

Figure 8 gives the recreation Engel curve. As in Figures 4–7, all four sets of estimates lie within the pointwise confidence intervals of the exact model estimates. However, as in the rent equation, the quadratic model is statistically rejected (see Table 2), and the Engel curve looks nonquadratic, particularly in the bottom decile, where the exact model estimate of the recreation Engel curve flattens out, while the quadratic estimate has a strong negative slope.

We now address the question of whether the complexity we observe in our estimated Engel curves is an artifact of our sample or of our choice of modeled expenditure categories. In particular, much of the nonquadratic behavior we observe is in the tails of the x distribution, where data are relatively sparse. In addition, many researchers of consumer demand exclude shelter expenditures (ie., rent) on the grounds that such expenditures cannot be adjusted quickly in the face of budget and price changes. A possible concern is whether the nonlinearity we observe

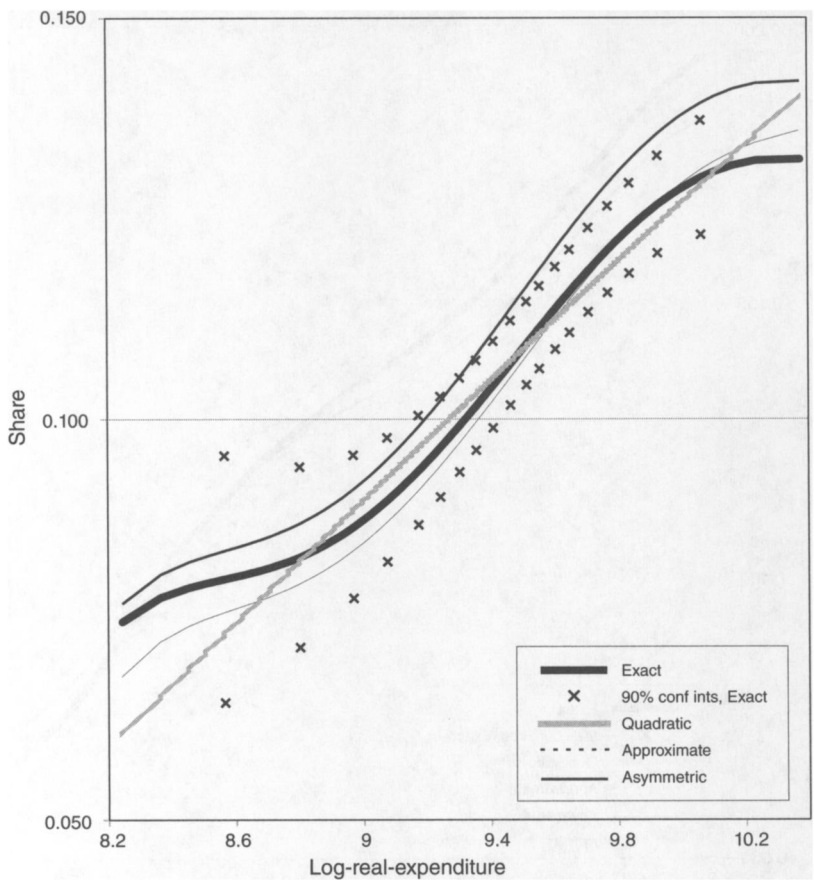


FIGURE 2. ESTIMATED FOOD-OUT SHARES

in the rent share artificially induces nonlinearity in other estimated budget share functions. We conduct three robustness exercises aimed at these issues. First, we estimated our EASI model using an extended sample of 18,600 households, which includes multiple-member households and households residing in Quebec. For this model, we included four additional demographic variables: the log of the number of household members, and dummies indicating single-parent households, childless-couple households, and households residing in Quebec. Second, we estimated our EASI model on the 8,846 households in our extended sample that did not have transfer income exceeding 10 percent of gross income and who did own a car (called the “nopoor” extended sample). Third, we estimated our EASI model on a subset of expenditure categories which excludes rent. For this model, we used households of all tenures, not just rental-tenure households, and used only those that did not have transfer income exceeding 10 percent of gross income and that did own a car. This yielded 32,399 observations available for an estimation sample (called the “norent nopoor” extended sample). For this sample, we include the number of rooms in the dwelling and its square as additional demographic characteristics. Detailed results for these models are available on request from the authors.

Figures 9 and 10 show the estimated rent and recreation Engel curves for various samples for the reference type defined above facing the base price vector, plus confidence intervals (shown

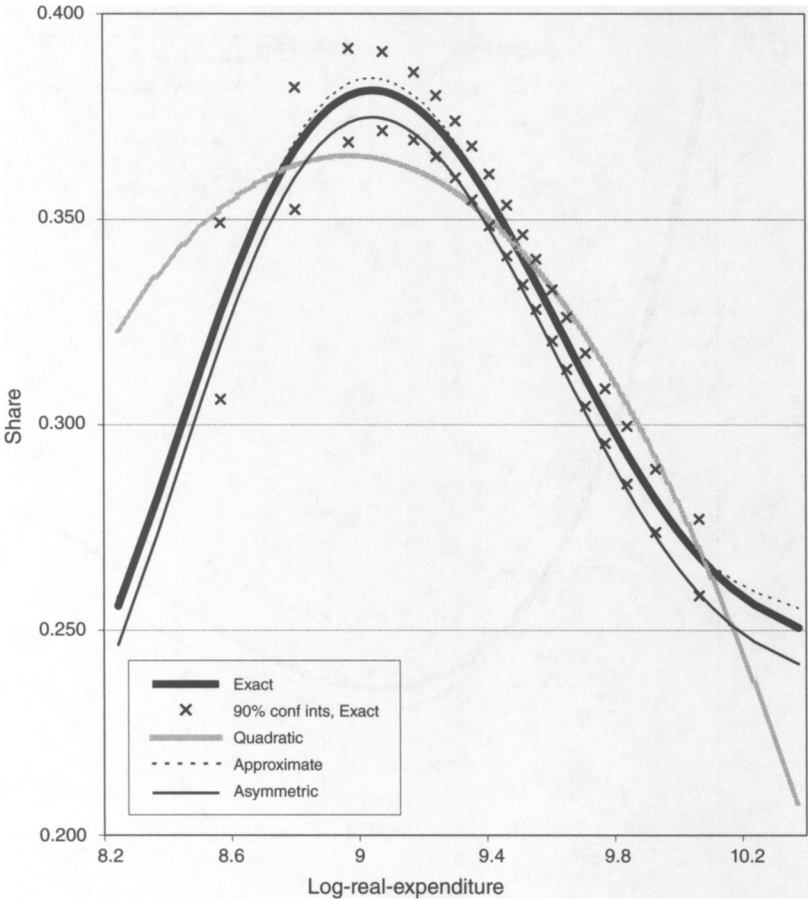


FIGURE 3. ESTIMATED RENT SHARES

with x's) for the Engel curves of the extended sample. Engel curves are evaluated at each fifth percentile of the single-person household real-expenditure distribution for the sample. Confidence intervals for other samples are similar in size, and are suppressed to reduce clutter. Examination of Figure 9 shows that the nonquadratic curvature we noted in Figure 3 for the rent Engel curve remains evident in the much larger extended sample (thin black line). Of course, the extended nopoor sample (dotted line) is uninformative at the bottom of the distribution because it excludes the poor, but it does reveal some departure from the quadratic model at the high end: at the ninety-seventh percentile of the nopoor distribution, the EASI Engel curve is 2 percentage points (and statistically significantly) higher than the quadratic model would indicate.

Figure 10 gives Engel curves for the baseline sample, the extended sample, the extended nopoor sample, and the extended nopoor norent sample. The figure shows that the curvature in the recreation Engel curve observed at the bottom end of the expenditure distribution in Figure 8 may have overstated the change in curvature for poor households. In particular, the extended sample (thin black line) does not reveal much nonquadratic curvature at the bottom of the distribution. Unsurprisingly, since the curvature was seen mainly at the bottom, the extended nopoor sample (dotted line) does not pick it up either. Also, in the larger sample at the upper end of the expenditure distribution, the two bunched recreation Engel curves lie within one another's 90

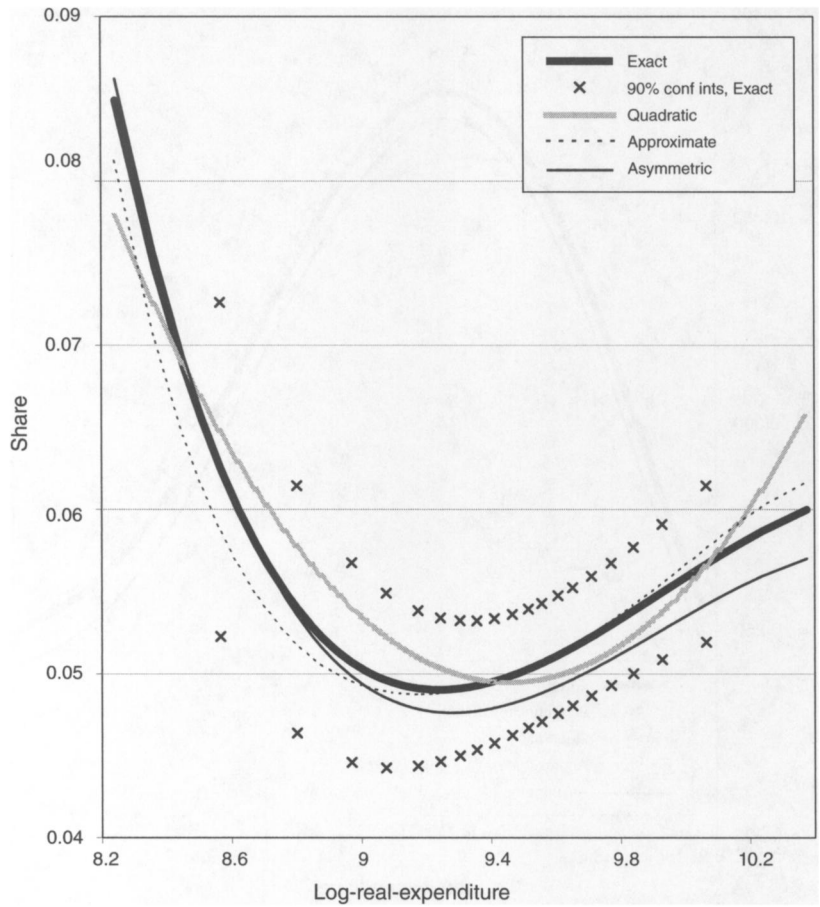


FIGURE 4. ESTIMATED HOUSEHOLD OPERATION SHARES

percent confidence intervals. Turning to the extended nopoor norent sample (thick grey line), wherein the rent equation is not modeled and a much larger number of observations is used, apart from a slight upturn at the very bottom of the distribution, we see little evidence of important nonquadratic curvature.

There are two important lessons that we draw from these figures. First, the SUR estimates of the approximate model do quite a good job of approximating the Engel curve estimates of the exact model, even when the underlying Engel curves are quite complex. Second, while the demand functions of some goods are close to linear or quadratic in log total expenditures, other goods such as rent and recreation are not quadratic. This implies a demand system rank (see Gorman 1981; Lewbel 1991) that is higher than three. Most past empirical studies have found rank to equal three, though Lewbel (2003) finds some empirical evidence for rank four. Figures 1–10 suggest that the reason previous studies failed to find ranks higher than three is because most of the departures from quadratic are either somewhat subtle, as in the household operation equation, or are concentrated in the tails of the expenditure distribution, as in the rent equation. Either way, the precision gained by detailed model specification and large sample sizes is needed to confirm the departure from quadratic Engel curves.

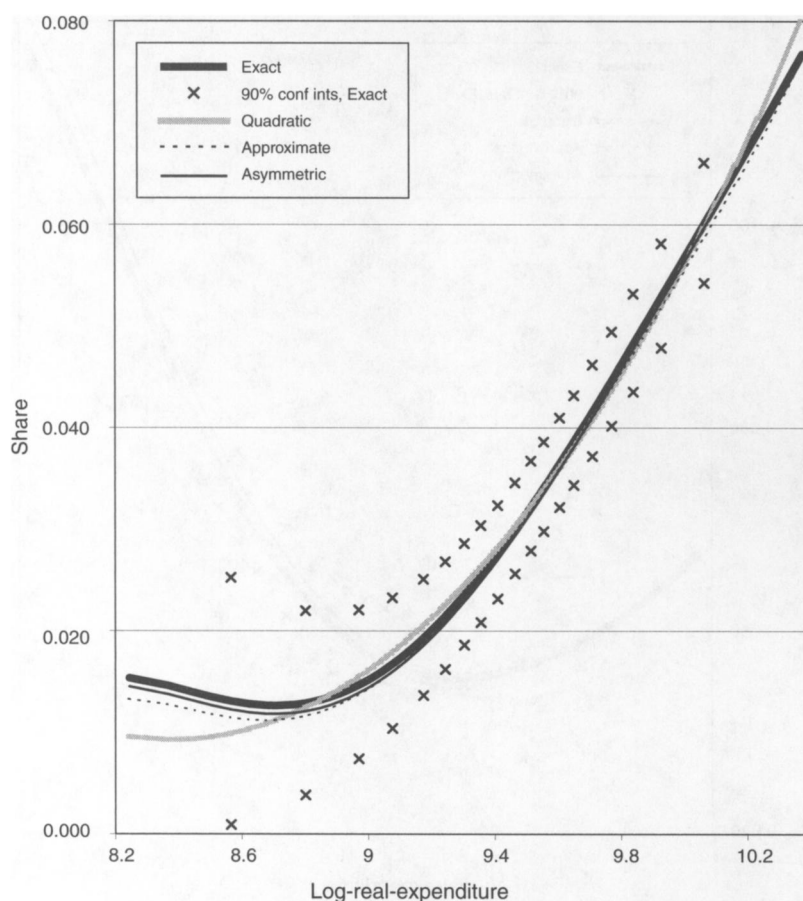


FIGURE 5. ESTIMATED HOUSEHOLD FURNISHING/EQUIPMENT SHARES

D. Estimated Price Effects

Prices in our model vary only by region and year, so despite having thousands of observations, we have only 48 different price regimes. However, these data still contain a very substantial amount of relative price variation, both because our sample is spread out over 28 years, and because Canada has historically had considerable regional price variation in some goods and services. As a result, we are able to obtain a reasonable amount of precision in many of our price effect estimates.

In our framework, price effects are most easily evaluated by looking at compensated budget-share semi-elasticities, compensated (good-specific) expenditure elasticities, or compensated quantity derivatives (aka, Slutsky terms). As shown by equation (12), compensated budget-share semi-elasticities with respect to prices are given by the matrix $\mathbf{\Upsilon} \equiv \sum_{l=0}^L \mathbf{A}_l \mathbf{z}_l + \mathbf{B} \mathbf{y}$, and so are affine in $\mathbf{z}$ and $\mathbf{y}$. Compensated (good-specific) expenditure elasticities with respect to prices are closely related and are given by $\mathbf{W}^{-1}(\mathbf{\Upsilon} + \mathbf{w}\mathbf{w}')$, where $\mathbf{W} = \text{diag}(\mathbf{w})$. The normalized¹ Slutsky matrix, $\mathbf{S}$, is related to the compensated semi-elasticity matrix, $\mathbf{\Upsilon}$, by $\mathbf{S} = \mathbf{\Upsilon} + \mathbf{w}\mathbf{w}' - \mathbf{W}$. Table 4

¹ The Slutsky matrix, $\bar{\mathbf{S}}$, is defined as the matrix of compensated quantity derivatives with respect to (unlogged) prices. The normalised Slutsky matrix, $\mathbf{S}$, (see Pollak and Wales 1992) normalizes the Slutsky matrix for prices and

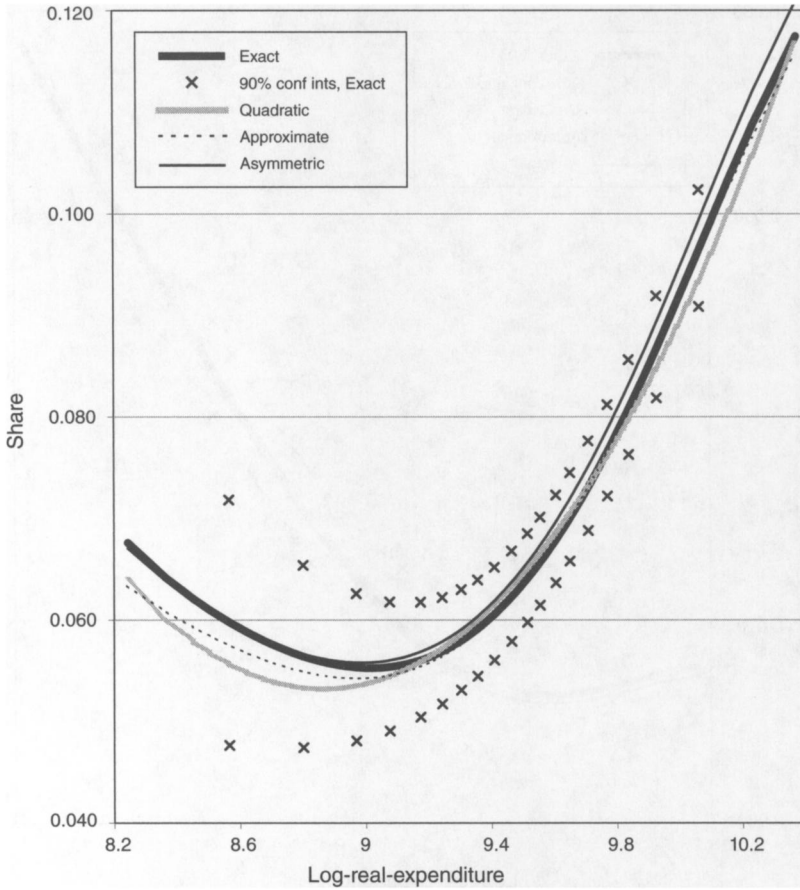


FIGURE 6. ESTIMATED CLOTHING SHARES

assesses all three of these measures of price effects. Since $y = x$ at the base price vector, and since $x = 0$ at median expenditure, this matrix is equal to A_0 at median expenditure for the reference type of person, a 40-year-old car-owning female in 1986 who did not receive much government transfer income. Table 4 presents summary estimated price effects, with asymptotic standard errors in *italics*. The rightmost block of Table 4 gives compensated price semi-elasticities for a reference person with median expenditure from the symmetry-restricted exact 3SLS estimates.

Consider, first, the matrix of compensated budget-share semi-elasticities for the reference person at median expenditure given by A_0 . Several of the own-price effects are large and statistically significant. The own-price compensated semi-elasticity for the rent budget share is 0.063, which implies that a rent price increase of 10 percent would be associated with a budget share 0.63 percentage points higher when expenditure is raised to equate utility with that in the initial situation. In contrast, if the recreation price rises by 10 percent, its budget share will be a little more than 1 percentage point lower when expenditure is raised to compensate for the loss in utility.

expenditure: $S = \bar{P}SP/x$, where $P = \text{diag}(\exp(p))$. Concavity of cost is necessary and sufficient for negative semidefiniteness of both the Slutsky matrix and the normalized Slutsky matrix.

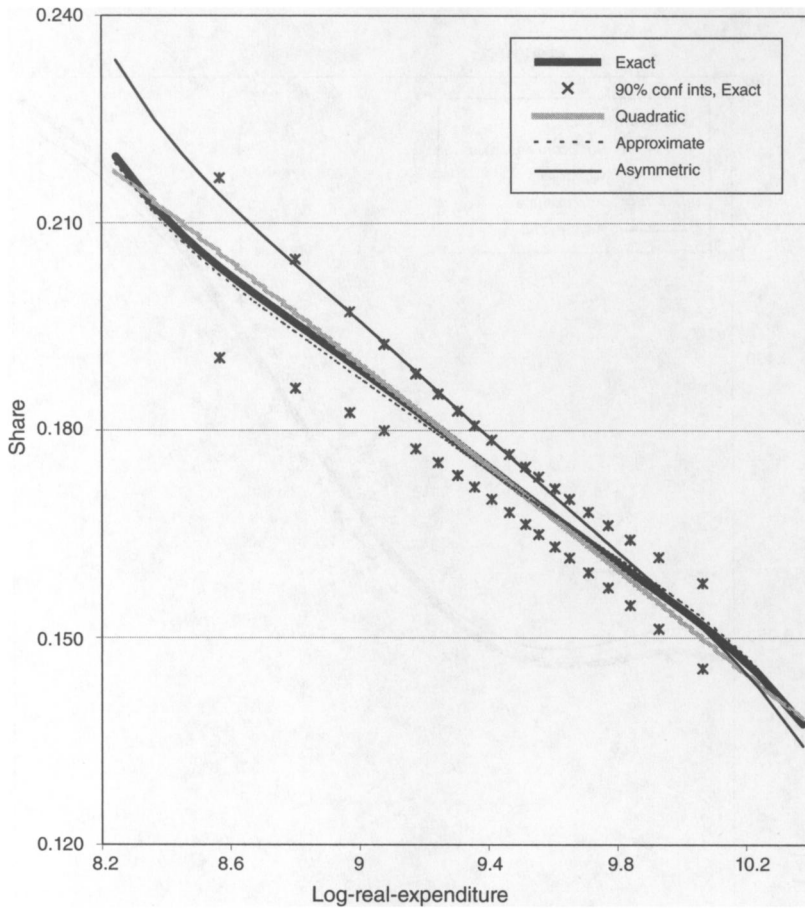


FIGURE 7. ESTIMATED TRANSPORTATION OPERATION SHARES

Several cross-price effects are also large and statistically significant, suggesting that substitution effects are important. For example, the clothing budget share compensated rent cross-price semi-elasticity is -0.066 , implying that an increase in the price of rent is associated with a significant decrease in the budget share for clothing, even after expenditure is raised to hold utility constant.

For some readers, expenditure elasticities may be more easily interpreted than budget-share semi-elasticities. In order to pin down these elasticities, we additionally need to specify the unobserved heterogeneity terms, which enter w (but don't enter Γ). For the reference type at median expenditures with $\varepsilon = 0$, we may calculate elasticities using A_0 and the level of the budget share function at median expenditure. The own-price expenditure elasticities are given in the third column of Table 4, with asymptotic standard errors computed via the delta method in *italics*. (The variance of these elasticities is driven by the variance of the own-price compensated semi-elasticities, Γ , and so reported standard errors are very close to those standard errors deflated by the appropriate budget share.) Unsurprisingly, only those equations with statistically significant own-price compensated budget-share semi-elasticities have statistically significant compensated expenditure elasticities. In particular, compensated rent expenditures have an elasticity of 0.528 and compensated transportation operation expenditures have an elasticity of 0.393. In contrast,

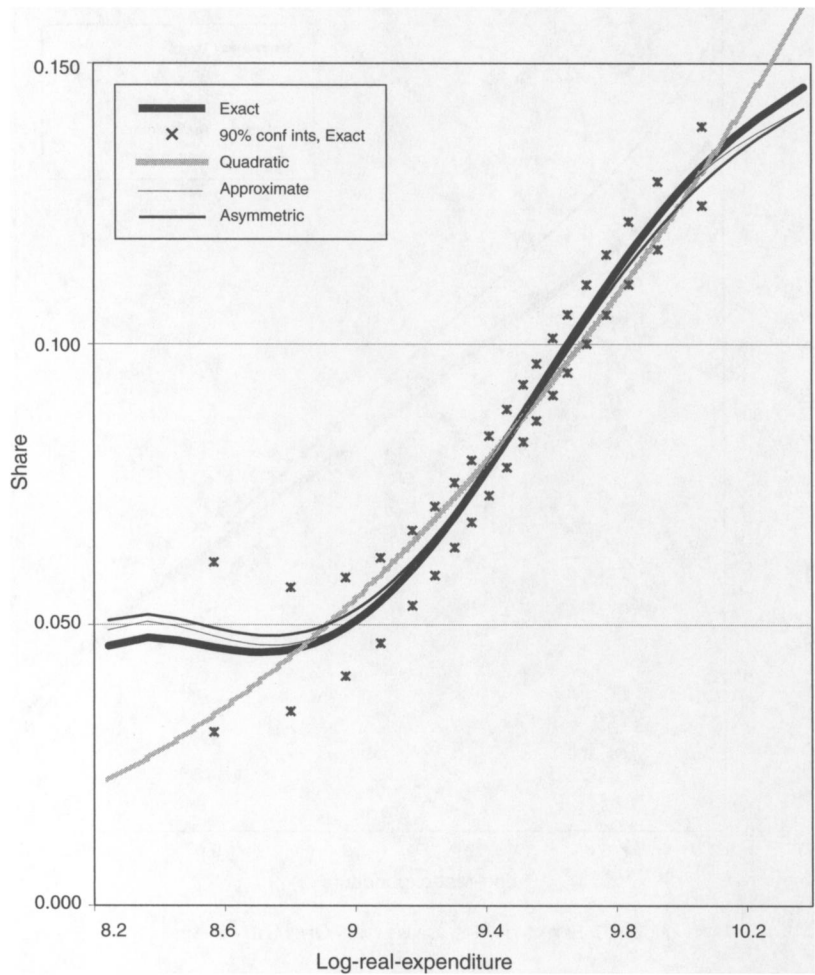


FIGURE 8. ESTIMATED RECREATION SHARES

compensated recreation expenditures are highly negatively elastic, with a marginally statistically significant own-price expenditure elasticity of -1.191 .

Although some of the own-price elasticities and semi-elasticities in Table 4 are statistically significantly positive, this does not imply that concavity (negative semi-definiteness) is violated. Concavity of cost is satisfied if and only if S is negative semi-definite (see, e.g., Pollak and Wales 1992). For the case where $\epsilon = 0_j$, the Slutsky matrix for the reference type with median expenditure facing base prices is fully specified by the matrix A_0 and the value of the Engel curve functions at median expenditure. The values of the own-price Slutsky terms are reported in the second column of Table 4, with asymptotic standard errors computed via the delta method in *italics*. A glance down this column reveals that the own-price Slutsky terms are all negative, and most are statistically significant. In addition, the Slutsky matrix evaluated at median expenditure for the reference type facing base prices is negative semidefinite, implying that the cost function is weakly concave at this point in the data.

The leftmost column of estimates in Table 4 contains the estimated own-price elements of B , which show the magnitudes of the interaction between own-prices and with log total expenditures.

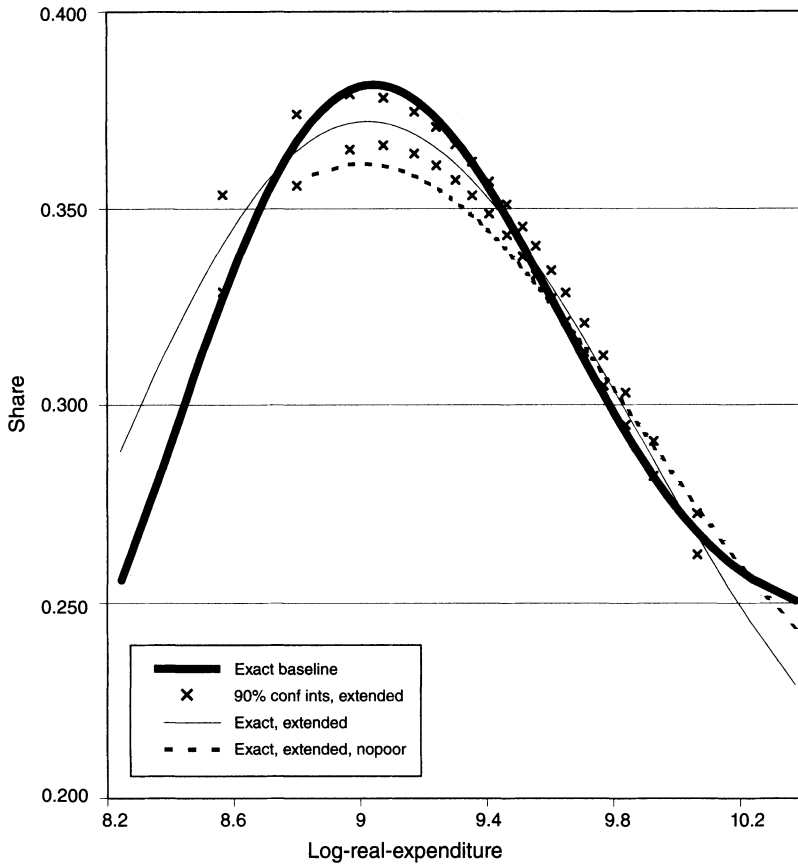


FIGURE 9. RENT SHARES: BASELINE VERSUS EXTENDED SAMPLE

These parameters allow us to assess whether or not compensated semi-elasticities are the same for rich and poor households. As noted in our discussion of Table 2, we cannot strongly reject the joint hypothesis that the entire matrix **B** is zero, so these results must be interpreted with caution. However, the estimated coefficient of the rent own-price compensated semi-elasticity on *y* is 0.088, and is by itself marginally statistically significant. Consider the comparison between the rent own-price compensated semi-elasticity for a reference person at the fifth percentile of expenditure ($x = -0.90$) versus that for such a person at the ninety-fifth percentile of expenditure ($x = 0.60$). As noted above, its value at the median expenditure of $x = 0$ is -0.066 . At the fifth percentile, its value is $-0.066 - (0.90 \times 0.088) = -0.016$, and is insignificantly different from zero. In contrast, at the ninety-fifth percentile, its value is $-0.066 + (0.60 \times 0.088) = 0.115$, and appears highly statistically significant. The corresponding own-price rent Slutsky terms are -0.236 at the fifth percentile and -0.080 at the ninety-fifth percentile, and both are statistically significantly negative. These results suggest that poor households substitute much more than do rich households in the face of an increase in the price of rent.

We draw three main conclusions from the analysis of price effects. First, we are able to obtain estimates of compensated elasticities, which, given symmetry, are second-order derivatives of the log-cost function and capture substitution effects. Uncompensated elasticities can also be calculated (see the online Appendix for details). These elasticity estimates suggest that some price effects, and therefore substitution effects, are large in magnitude. Our second conclusion is that the rationality

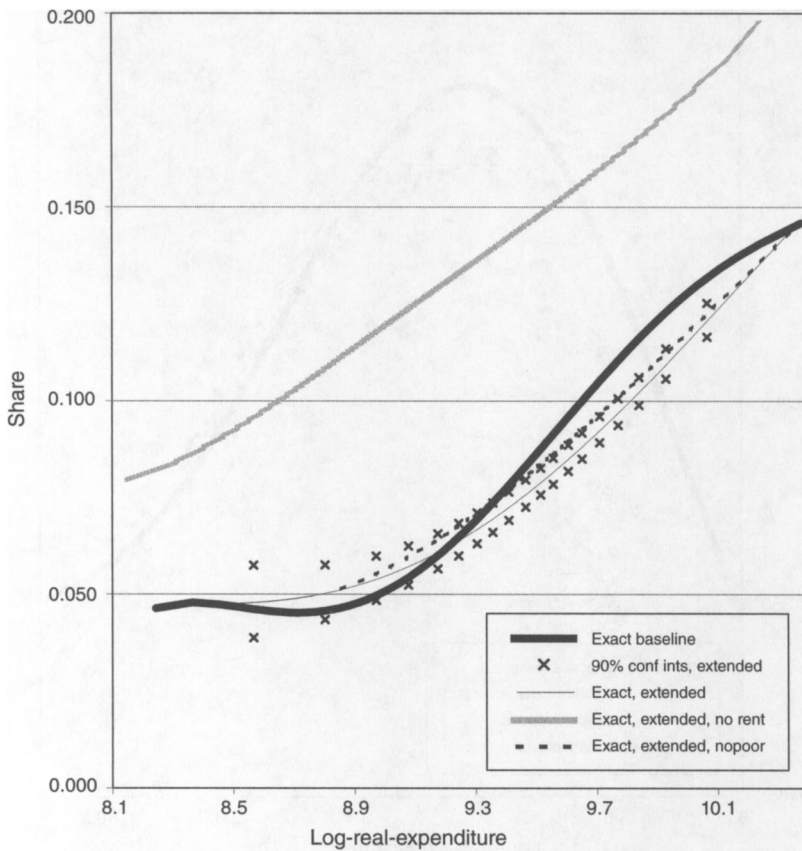


FIGURE 10. RECREATION SHARES, BASELINE VERSUS EXTENDED SAMPLE

restriction of concavity is not violated, at least for the reference type. Third, there is some evidence that substitution effects are different for rich and poor households and, specifically, that poor households substitute much more in the face of rental price increases than do rich households.

E. Consumer Surplus Estimates

We assess the economic significance of our models with a cost-of-living experiment. In Canada, rent is not subject to sales taxes, which typically amount to 15 percent for goods such as food-out and clothing. Consider the cost-of-living index associated with subjecting rent to a 15 percent sales tax for people facing the base price vector, so that $\mathbf{p}_0 = \mathbf{0}_J$ and $\mathbf{p}_1 = [0 \ 0 \ \ln 0.15 \ 0 \ 0 \ 0 \ 0 \ 0]$:

$$\begin{aligned} C(\mathbf{p}_1, u, \mathbf{z}, \boldsymbol{\epsilon}) - C(\mathbf{p}_0, u, \mathbf{z}, \boldsymbol{\epsilon}) &= \ln 0.15 w^{\text{rent}} + \ln 0.15^2 \left(\sum_{l=0}^L z_l a_l^{\text{rent}, \text{rent}} + b^{\text{rent}, \text{rent}} y \right) / 2 \\ &= \ln 0.15 w^{\text{rent}} + \ln 0.15^2 \left(\sum_{l=0}^L z_l a_l^{\text{rent}, \text{rent}} + b^{\text{rent}, \text{rent}} x \right) / 2, \end{aligned}$$

where $a_l^{\text{rent}, \text{rent}}$ and $b^{\text{rent}, \text{rent}}$ are the rent own-price elements of $\mathbf{A}_l$ and $\mathbf{B}$. We choose $\mathbf{0}_J$ as the comparison price vector because at this price vector, $y = x$, and as a consequence only one budget share and six parameters are needed to estimate the cost-of-living index. Here, unobserved

TABLE 4—COMPENSATED PRICE EFFECTS, EVALUATED FOR REFERENCE TYPE WITH
MEDIAN EXPENDITURE AT BASE PRICES

	Own-price B element	Own-price Slutsky terms	Own-price Quant elast	Budget-share semi-elasticities							
				Food in	Food out	Rent	HH oper	HH furneq	Clothing	Tran oper	Recr
Food at home	0.047 (0.063)	−0.137 (0.045)	−0.064 (0.347)	−0.025 (0.045)							
Food out	−0.009 (0.068)	−0.120 (0.037)	−0.126 (0.346)	0.051 (0.033)	−0.025 (0.037)						
Rent	0.088 (0.047)	−0.164 (0.029)	0.528 (0.083)	−0.026 (0.021)	0.037 (0.021)	0.063 (0.029)					
Household oper	0.044 (0.039)	−0.035 (0.026)	0.292 (0.517)	0.026 (0.026)	−0.002 (0.022)	−0.002 (0.014)	0.012 (0.026)				
Household furneq	0.037 (0.114)	−0.034 (0.058)	−0.150 (1.937)	0.110 (0.035)	−0.043 (0.032)	0.028 (0.021)	−0.018 (0.026)	−0.005 (0.058)			
Clothing	−0.050 (0.060)	−0.066 (0.046)	−0.063 (0.734)	−0.050 (0.030)	−0.001 (0.032)	−0.066 (0.020)	−0.025 (0.024)	−0.023 (0.038)	−0.008 (0.046)		
Transport oper	−0.006 (0.035)	−0.105 (0.020)	0.393 (0.115)	−0.023 (0.021)	−0.030 (0.019)	−0.019 (0.016)	−0.013 (0.017)	−0.034 (0.021)	0.053 (0.019)	0.038 (0.020)	
Recreation	−0.030 (0.093)	−0.182 (0.053)	−1.191 (0.635)	−0.044 (0.034)	0.032 (0.032)	−0.021 (0.023)	0.034 (0.025)	−0.010 (0.045)	0.087 (0.037)	0.033 (0.022)	−0.106 (0.053)

Note: Standard errors in parentheses.

heterogeneity enters only through the level effect on w^{rent} . We can think of this cost-of-living index as comprising two effects: a first-order effect which is driven by expenditure shares and which incorporates unobserved heterogeneity; and a second-order effect which captures substitution effects. Traditional consumer demand analysis, which ignores unobserved heterogeneity, would accommodate both first- and second-order effects, but would use $\hat{w}^{rent}$, which contains no “error term,” rather than w^{rent} , which contains an unobserved preference heterogeneity component. In contrast, traditional nonparametric approaches to the cost of living would use only the first-order term which accommodates unobserved heterogeneity, but would not incorporate the second-order term which captures substitution effects. Our model combines the advantages of both approaches.

Figure 11 shows the estimated values of the cost-of-living index for each household facing the base price vector in our baseline sample incorporating unobserved heterogeneity with empty circles, and shows estimated values for each household with unobserved heterogeneity set to zero ($\epsilon = 0$) using filled circles. In addition, the second-order component capturing substitution effects is shown with filled squares. The reason that the filled circles and filled squares do not each lie on a single line is that variation in demographic characteristics $\mathbf{z}$ across households affects the surplus measures.

The underlying Engel curve is visible in the estimates which zero out ϵ , but is largely obscured when this unobserved heterogeneity is taken into account. Failure to account for unobserved heterogeneity leads to the erroneous impression that most of the variation across individuals in the cost-of-living impact of a large rent increase is related to expenditure, and only a little is related to other characteristics. The more refined picture is that most of the variation in the impact on cost of living is attributable to unobserved characteristics. Even in a model as rich as ours, with many hundreds of parameters, most of the variation in demand is due to unobserved characteristics, and this is reflected in the variation in cost-of-living responses. As noted earlier, our parameter estimates would be little changed if some or all of the errors were interpreted as ordinary modeling error rather than preference heterogeneity. In this case, the cost of living impacts would lie somewhere between the filled and empty circles in Figure 11.

Figure 11 also shows the need for highly flexible Engel curves. The first-order term in the consumer surplus calculation is driven by the Engel curve, and as Figure 3 shows, even a quadratic provides a poor approximation to the rent expenditure share equation, so demand systems that allow for only linear or quadratic Engel curves can make substantial errors in policy analyses. These errors would be magnified in a policy experiment that more directly affected the distribution of total expenditures, such as a change in the progressivity of income taxes.

The second-order terms above capture substitution effects across expenditure share equations. These effects are not large in this experiment, but they do have a pronounced pattern, as shown by the filled squares in Figure 11. If consumers substitute greatly in the face of price increases, then the second-order terms will be large and negative; if they substitute little, then the second-order terms will be large and positive. Since the poor substitute more than the rich, second-order effects are positively related to expenditure, so ignoring them would result in underestimating the cost-of-living impact for rich households and overestimating the impact for poor households. The magnitude of the second-order term is about -0.1 percentage points for households at the fifth percentile (on a total impact of about 6 percentage points), and its magnitude is about $+0.1$ percentage points for households at the ninety-fifth percentile (on a total impact of about 4 percentage points).

As noted above, the treatment of ϵ as unobserved preference heterogeneity parameters rather than as measurement error affects consumer surplus measures greatly. However, it turns out not to affect estimated parameters of the model very much. We reestimated the EASI model assuming ϵ is measurement error via an iterative procedure. First, we estimate the exact EASI model, and then compute $y(\mathbf{p}'(\mathbf{w} - \mathbf{e}))$ as in Table 2, which is identical to y except that it has $\mathbf{p}'\mathbf{e}$ subtracted from the numerator in the definition of y given in equation (8). Then, we estimate the approximate EASI model using $y(\mathbf{p}'(\mathbf{w} - \mathbf{e}))$ as $\tilde{y}$, recompute $y(\mathbf{p}'(\mathbf{w} - \mathbf{e}))$, reestimate the approximate EASI model, and iterate to convergence (four iterations). The resulting estimated parameter vector is very close to the exact EASI estimated parameter vector. The maximum absolute distance between parameters of 0.1 standard error (using the exact EASI estimated standard errors), and the average absolute distance is 0.005 standard error. In terms of an economic measure of difference, using the “measurement error” version of the model to assess cost-of-living impacts as in Figure 11 yields estimates visually identical to the filled black circles in the figure. The maximum absolute distance between the filled black circles in the figure and the “measurement error” version of cost-of-living estimates is 0.01 percentage points, which is negligible in comparison to the estimated values of around 5 percentage points.

III. Conclusions

We invent *implicit Marshallian demand* systems. These are the compensated budget share functions (Hicks demands) associated with cost functions having the property that utility can be represented by a simple function of observables, including budget shares themselves. We show that simple implicit Marshallian demand systems allow us to solve many problems facing consumer demand analysis, including the incorporation of unobserved preference heterogeneity and allowing for flexibility in Engel curves.

We provide the *Exact Affine Stone Index* (EASI) implicit Marshallian demand system, in which utility is ordinaly equivalent to an affine function of the log of expenditure deflated by the Stone Index. This EASI demand system is as flexible in price responses, as close to linear in parameters, and as easy to estimate as the Almost Ideal Demand (AID) system. But, in contrast to the AID system, the EASI demand system also allows for flexible interactions between prices and expenditures, permits almost any functional form for Engel curves, and allows error terms in the model to correspond to unobserved preference heterogeneity random utility parameters. Demand functions may be estimated by 3SLS or GMM, and, like the AID

system, an approximate model can be estimated by linear regression. We estimate the model, and find some significant departures from linear and quadratic demands. We also find that allowing errors to equal unobserved preference heterogeneity has little effect on model parameter estimates, but substantially affects social welfare calculations such as the cost-of-living impacts of price changes.

APPENDIX A: EXACT STONE INDEX (ESI) DEMANDS²

Ignore preference heterogeneity for now; that is, consider $\mathbf{z}$ and $\boldsymbol{\varepsilon}$ as fixed. We will reintroduce them later. Define preferences to be *regular* if they can be represented by a log cost function $C(\mathbf{p}, u)$, where $\exp[C(\mathbf{p}, u)]$ is concave, increasing, differentiable, and homogeneous of degree 1 in (unlogged) prices $\exp(\mathbf{p})$ and is monotonically increasing and differentiable in u . For a given $C(\mathbf{p}, u)$ function, it may be necessary to restrict the domain of $\mathbf{p}, u$ to values where $C(\mathbf{p}, u)$ is regular. Shephard's lemma relates Hicksian (compensated) budget shares to regular cost functions by

$$\mathbf{w} = \boldsymbol{\omega}(\mathbf{p}, u) = \nabla_{\mathbf{p}} C(\mathbf{p}, u).$$

Consider log Stone index deflated expenditures $x - \mathbf{p}'\mathbf{w}$. Suppose preferences are represented by a cost function $C(\mathbf{p}, u)$ that makes $u = y$, where implicit utility $y = x - \mathbf{p}'\mathbf{w}$, which is the log of Stone index deflated expenditures. We call this an Exact Stone Index (ESI) cost function. If we have an ESI cost function, then we can substitute out u in the Hicks demand functions $\mathbf{w} = \boldsymbol{\omega}(\mathbf{p}, u)$ to obtain $\mathbf{w} = \boldsymbol{\omega}(\mathbf{p}, x - \mathbf{p}'\mathbf{w})$. The name Exact Stone Index is in contrast with the approximate Almost Ideal Demand System, which uses $y = x - \mathbf{p}'\mathbf{w}$ as an approximation to deflating x by a certain quadratic function of $\mathbf{p}$. In an ESI cost function, the Stone index is not an approximation to some true deflator. Instead, the Stone index is the exact correct deflator for x .

Given an ESI cost function, $\mathbf{w} = \boldsymbol{\omega}(\mathbf{p}, x - \mathbf{p}'\mathbf{w})$ is an example of an implicit Marshallian demand functions. Using $x = C(\mathbf{p}, u)$ and Shephard's lemma, we obtain $u = x - \mathbf{p}'\mathbf{w}$, and therefore have an ESI cost function, if and only if

$$(A1) \quad u = C(\mathbf{p}, u) - \mathbf{p}'[\nabla_{\mathbf{p}} C(\mathbf{p}, u)].$$

Theorem 1 characterizes the solutions to this equation.

THEOREM 1: Define $d(\mathbf{p}, u) \equiv C(\mathbf{p}, u) - u$ and assume $C(\mathbf{p}, u)$ is a regular cost function. Then $C(\mathbf{p}, u)$ is an ESI cost function if and only if $d(\lambda\mathbf{p} + \mathbf{1}_J\kappa, u) = \lambda d(\mathbf{p}, u) + \kappa$ for any scalars $\lambda > 0$ and κ .

PROOF OF THEOREM 1: $d(\lambda\mathbf{p}, u) = \lambda d(\mathbf{p}, u)$ is equivalent to $d(\mathbf{p}, u)$ linearly homogeneous in $\mathbf{p}$, and $d(\mathbf{p} + \mathbf{1}_J\kappa, u) = d(\mathbf{p}, u) + \kappa$ is equivalent to $\exp[d(\mathbf{p}, u)]$ linearly homogeneous in $\exp(\mathbf{p})$. The latter condition is required by cost function regularity. For the former, given equation (A1) we have $C(\mathbf{p}, u) - u = \mathbf{p}'[\nabla_{\mathbf{p}} C(\mathbf{p}, u)]$. This is equivalent to $d(\mathbf{p}, u) = \mathbf{p}'[\nabla_{\mathbf{p}} d(\mathbf{p}, u)]$, which by the Euler Theorem shows that $d(\mathbf{p}, u)$ is linearly homogeneous in $\mathbf{p}$.

² The following materials appear in the online Appendix: I. General Cost Functions for Implicit Marshallian Demands; II. Generalized EASI Cost Functions; III. Coherency, Invertibility, and the Distribution of Errors; IV. The Rank of EASI Demand Systems; V. Shape Invariance and Equivalence Scales; VI. Closure under Unit Scaling; VII. Marshallian Elasticity Calculations.

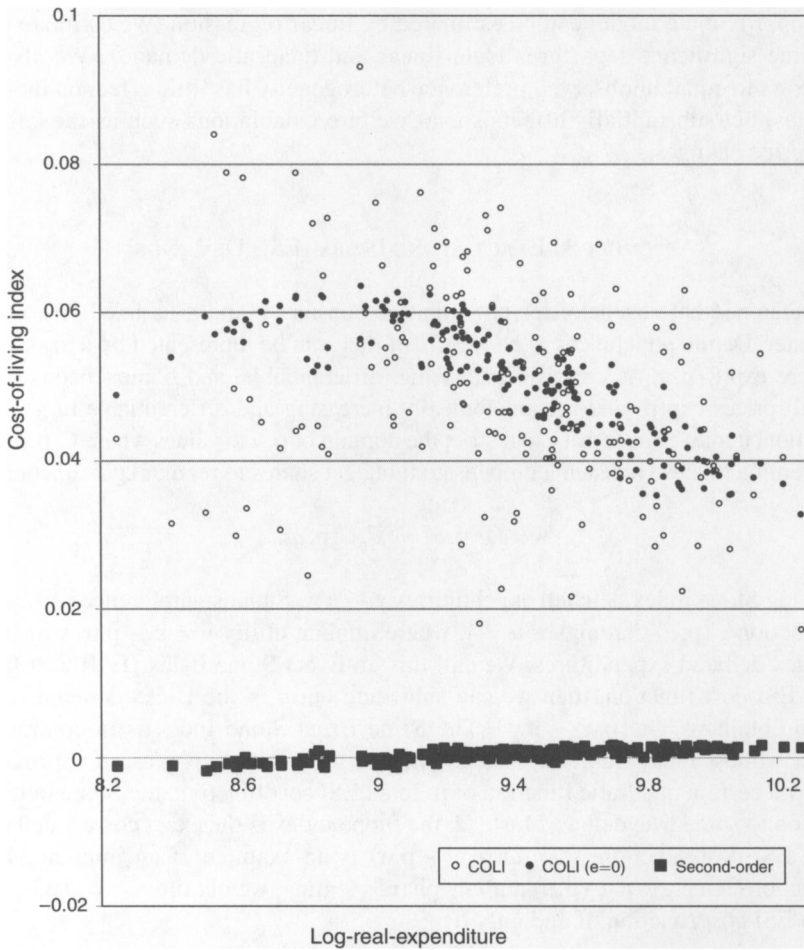


FIGURE 11. COST-OF-LIVING EXPERIMENT: 15% TAX ON RENT

Theorem 1 can be equivalently stated as the requirement that $\exp[d(\mathbf{p}, u)]$ be linearly homogeneous in $\exp(\mathbf{p})$, which follows from ordinary cost function homogeneity, and also that $d(\mathbf{p}, u)$ be linearly homogeneous in $\mathbf{p}$, which makes $u = x - \mathbf{p}'\mathbf{w}$. A function $d(\mathbf{p})$ is defined to be, “translatable of degree c ” if $d(\mathbf{p} + \mathbf{1}, \kappa) = d(\mathbf{p}, u) + c\kappa$ (see, e.g., Blackorby and Donaldson 1980), so another way to state Theorem 1 is that $C(\mathbf{p}, u)$ is an ESI cost function if and only if $C(\mathbf{p}, u) - u$ is both linearly homogeneous in $\mathbf{p}$ and translatable of degree 1 in $\mathbf{p}$.

By ordinality of utility, instead of $x - \mathbf{p}'\mathbf{w} = u$, we could have defined Stone Index exactness by $x - \mathbf{p}'\mathbf{w} = h(u)$ for any strictly monotonically increasing function h , and Theorem 1 would then hold with $d(\mathbf{p}, u)$ defined by $C(\mathbf{p}, u) - h(u)$. There is no gain in generality from doing so, because, again by ordinality, $C(\mathbf{p}, u) - h(u)$ has the same indifference curves, and hence the same Marshallian and implicit Marshallian demand functions, as $C(\mathbf{p}, h^{-1}(u)) - u = \tilde{C}(\mathbf{p}, u) - u$. In this representation of the cost function (i.e., in this cardinalization of utility), we then obtain $x - \mathbf{p}'\mathbf{w} = u$ as before.

The following corollary and theorem illustrates the restrictiveness of ESI demands.

COROLLARY 1: Assume $C(\mathbf{p}, u)$ is an ESI cost function. Then Hicksian demands $\omega(\mathbf{p}, u) = \nabla_{\mathbf{p}} C(\mathbf{p}, u)$ are homogeneous of degree 0 in $\mathbf{p}$ and in $\exp(\mathbf{p})$.

PROOF OF COROLLARY 1: Homogeneity of degree 0 in $\exp(\mathbf{p})$ follows from ordinary cost function regularity, and homogeneity of degree 0 in $\mathbf{p}$ follows from the property that the derivative of a linearly homogeneous function, in this case $d(\mathbf{p}, u)$, must be homogeneous of degree 0.

Cost function regularity implies that Hicksian demands do not change when all prices are scaled by a constant factor. Corollary 1 shows that ESI requires that Hicksian demands also not change when all prices undergo the same power transformation, for example, when all prices are squared, and also rules out Hicksian demands that are linear in $\mathbf{p}$. Relative prices can change dramatically when all prices are squared, so this additional homogeneity condition binds in economically implausible ways.

THEOREM 2: Consider the case where $J = 2$. Then, $d(\lambda p_1 + \kappa, \lambda p_2 + \kappa, u) = \lambda d(p_1, p_2, u) + \kappa$ for any scalars $\lambda > 0$ and κ if and only if $d(p_1, p_2, u) = m(u)p_1 + [1 - m(u)]p_2$ for some function $m(u)$.

PROOF OF THEOREM 2: Since $d(\lambda p_1 + \kappa, \lambda p_2 + \kappa, u) = \lambda d(p_1, p_2, u) + \kappa$ holds for any λ and κ , let $\lambda = 1/(p_2 - p_1)$ and $\kappa = -p_1/(p_2 - p_1)$ to obtain $d(0, 1, u) = [d(p_1, p_2, u)/(p_2 - p_1)] - [p_1/(p_2 - p_1)]$. The theorem holds with $m(u) = 1 - d(0, 1, u)$.

Theorem 2 shows that with $J = 2$ goods, ESI cost functions must have the linear in $\mathbf{p}$ form $C(\mathbf{p}, u) = u + \mathbf{p}'\mathbf{m}(u)$ where $\mathbf{m}(u)$ is a J -vector of functions that satisfy $\mathbf{m}(u)' \mathbf{1}_J = 1$. This is an ESI cost function for any J , but has the unattractive feature that it has Hicksian budget shares $\mathbf{m}(u)$ that are independent of $\mathbf{p}$.

ESI cost functions having more than $J = 2$ goods can be nonlinear in $\mathbf{p}$. One example is

$$C(\mathbf{p}, u) = u + \mathbf{p}'\mathbf{m}(u) + [\mathbf{p}'\mathbf{M}(u)\mathbf{p}]^{1/2},$$

where $\mathbf{m}(u)$ is as above and $\mathbf{M}(u)$ is a J by J symmetric matrix-valued function of u with $\mathbf{M}(u)\mathbf{1}_J = \mathbf{0}_J$. As this example shows, ESI log cost functions can be nonlinear in log prices; however, by Corollary 1, ESI log cost functions cannot be quadratic in log prices, and therefore ESI budget shares cannot be linear in log prices.

APPENDIX B: EXACT AFFINE STONE INDEX (EASI) DEMANDS

As shown in the previous section, ESI Hicks and implicit Marshallian budget shares must possess the unattractive feature of not changing when all prices are squared, and also must either be independent of $\mathbf{p}$ or nonlinear in $\mathbf{p}$. We can avoid these drawbacks by generalizing the expression for real expenditures, thereby relaxing the homogeneity restrictions required by ESI demands. Specifically, instead of imposing the ESI restriction that some cardinalization of u equals $x - \mathbf{p}'\mathbf{w}$, define EASI cost functions to be cost functions that have the property that a cardinalization of u exists that equals an affine transformation of $x - \mathbf{p}'\mathbf{w}$. In this cardinalization, $u = y$, where implicit utility y is defined by $y = (x - \mathbf{p}'\mathbf{w})\tilde{s}(\mathbf{p}) - \tilde{t}(\mathbf{p})$ for some functions $\tilde{s}(\mathbf{p})$ and $\tilde{t}(\mathbf{p})$. For reasons that will be clear shortly, it is more convenient to express this relationship as

$$(B1) \quad u = y = \frac{x - \mathbf{p}'\mathbf{w} - t(\mathbf{p})}{1 + s(\mathbf{p})}$$

for some functions $s(\mathbf{p})$ and $t(\mathbf{p})$. We choose affine transforms for their simplicity; in particular, for now we wish to avoid having to solve nonlinear transformations to obtain cost functions (online Appendix I discusses more general implicit Marshallian demand functions that may require nonlinear transformation).

Equation (18) implies

$$x - u - \mathbf{p}'\mathbf{w} = t(\mathbf{p}) + s(\mathbf{p})u,$$

and therefore any EASI cost function $C(\mathbf{p}, u)$ must satisfy

$$(B2) \quad C(\mathbf{p}, u) - u - \mathbf{p}'[\nabla_{\mathbf{p}}C(\mathbf{p}, u)] = t(\mathbf{p}) + s(\mathbf{p})u.$$

The following two theorems characterize EASI cost functions, and hence the solutions to equation (B2), and provide a convenient way to construct such functions.

THEOREM 3: Assume $C(\mathbf{p}, u)$ is regular. Let $d(\mathbf{p}, u) \equiv C(\mathbf{p}, u) - u$. There exists functions $t(\mathbf{p})$ and $s(\mathbf{p})$ that make equation (B1) hold, and hence make $C(\mathbf{p}, u)$ an EASI cost function, if and only if $d(\mathbf{p} + \mathbf{1}_J\kappa, u) = d(\mathbf{p}, u) + \kappa$ and $\nabla_u^2 d(\lambda\mathbf{p}, u) = \lambda\nabla_u^2 d(\mathbf{p}, u)$ for any scalars $\lambda > 0$ and κ . When equation (B1) holds, the functions $t(\mathbf{p})$ and $s(\mathbf{p})$ satisfy $t(\mathbf{p} + \mathbf{1}_J\kappa) = t(\mathbf{p}) + \kappa$ and $s(\mathbf{p} + \mathbf{1}_J\kappa) = s(\mathbf{p}) + \kappa$.

PROOF OF THEOREM 3: The condition that $d(\mathbf{p} + \mathbf{1}_J\kappa, u) = d(\mathbf{p}, u) + \kappa$ is equivalent to $\exp[d(\mathbf{p}, u)]$ linearly homogeneous in $\exp(\mathbf{p})$, which is required by cost function regularity. Equation (B2) is equivalent to $d(\mathbf{p}, u) - t(\mathbf{p}) - s(\mathbf{p})u = \mathbf{p}'[\nabla_{\mathbf{p}}d(\mathbf{p}, u)]$. Taking derivatives of this expression with respect to u gives $\nabla_u d(\mathbf{p}, u) - s(\mathbf{p}) = \mathbf{p}'[\nabla_{\mathbf{p}}\nabla_u d(\mathbf{p}, u)]$, so $\nabla_u^2 d(\mathbf{p}, u) = \mathbf{p}'[\nabla_{\mathbf{p}}\nabla_u^2 d(\mathbf{p}, u)]$, which by the Euler Theorem shows that $\nabla_u^2 d(\mathbf{p}, u)$ is linearly homogeneous in $\mathbf{p}$, that is, $\nabla_u^2 d(\lambda\mathbf{p}, u) = \lambda\nabla_u^2 d(\mathbf{p}, u)$. In the other direction, if $\nabla_u^2 d(\mathbf{p}, u)$ is linearly homogeneous then $\nabla_u^2 d(\mathbf{p}, u) = \mathbf{p}'[\nabla_{\mathbf{p}}\nabla_u^2 d(\mathbf{p}, u)]$. Integrate this equation twice with respect to u (defining $s(\mathbf{p})$ to be the constant of integration for the first integration and $t(\mathbf{p})$ to be the constant of integration for the second integration) and continue following the steps above in reverse to obtain equation (B2). Given exactness, equation (B2) implies $t(\mathbf{p}) + s(\mathbf{p})u = u - C(\mathbf{p}, u) + \mathbf{p}'\omega(\mathbf{p}, u)$. Ordinary homogeneity of cost functions and Hicksian demands requires that $\exp[u - C(\mathbf{p}, u) + \mathbf{p}'\omega(\mathbf{p}, u)]$ be linearly homogeneous in $\exp(\mathbf{p})$, so $t(\mathbf{p}) + s(\mathbf{p})u$ must be linearly homogeneous in $\exp(\mathbf{p})$. Since u can vary holding $\mathbf{p}$ fixed, this requires that $t(\mathbf{p})$ and $s(\mathbf{p})$ must each be linearly homogeneous in $\exp(\mathbf{p})$, and therefore that $t(\mathbf{p} + \mathbf{1}_J\kappa) = t(\mathbf{p}) + \kappa$ and $s(\mathbf{p} + \mathbf{1}_J\kappa) = s(\mathbf{p}) + \kappa$.

THEOREM 4: Assume the function $\bar{C}(\mathbf{p}, u)$ satisfies the homogeneity conditions required to be an ESI log cost function as given by Theorem 1. Assume $C(\mathbf{p}, u) = \bar{C}(\mathbf{p}, u) + T(\mathbf{p}) + S(\mathbf{p})u$ is a regular log cost function. Then $C(\mathbf{p}, u)$ is an EASI log cost function, so equation (B1) holds, with $t(\mathbf{p}) = T(\mathbf{p}) - \mathbf{p}'[\nabla_{\mathbf{p}}T(\mathbf{p})]$ and $s(\mathbf{p}) = S(\mathbf{p}) - \mathbf{p}'[\nabla_{\mathbf{p}}S(\mathbf{p})]$.

PROOF OF THEOREM 4: $C(\mathbf{p}, u) = \bar{C}(\mathbf{p}, u) + T(\mathbf{p}) + S(\mathbf{p})u$, so

$$\begin{aligned} C(\mathbf{p}, u) - \mathbf{p}'[\nabla_{\mathbf{p}}C(\mathbf{p}, u)] &= \bar{C}(\mathbf{p}, u) - \mathbf{p}'[\nabla_{\mathbf{p}}\bar{C}(\mathbf{p}, u)] + T(\mathbf{p}) - \mathbf{p}'[\nabla_{\mathbf{p}}T(\mathbf{p})] \\ &\quad + S(\mathbf{p})u - \mathbf{p}'[\nabla_{\mathbf{p}}S(\mathbf{p})]u \\ &= u + T(\mathbf{p}) - \mathbf{p}'[\nabla_{\mathbf{p}}T(\mathbf{p})] + S(\mathbf{p})u - \mathbf{p}'[\nabla_{\mathbf{p}}S(\mathbf{p})]u, \end{aligned}$$

where the right side of the second equality follows from Stone index exactness for $\bar{C}(\mathbf{p}, u)$ from Theorem 1. Comparing this equation to (B2) proves the result.

Now consider how observed variables $\mathbf{z}$ that affect preferences and how random unobserved preference heterogeneity parameters $\boldsymbol{\varepsilon}$ can be incorporated into the ESI and EASI cost functions. These cost functions are formulated above in terms of the functions $\bar{C}(\mathbf{p}, u)$, $T(\mathbf{p})$ and $S(\mathbf{p})$. Incorporation of observed heterogeneity is completely general in that there are no restrictions on how observed characteristics $\mathbf{z}$ enter the ESI cost function $\bar{C}(\mathbf{p}, u, \mathbf{z})$ or the price functions $T(\mathbf{p}, \mathbf{z})$ and $S(\mathbf{p}, \mathbf{z})$. However, the incorporation of unobserved heterogeneity is restricted by the fact that implicit utility y must be a function of observables only. This can be accomplished by letting $\bar{C}$ depend arbitrarily on $\boldsymbol{\varepsilon}$ and having the price functions $T(\mathbf{p}, \mathbf{z})$ and $S(\mathbf{p}, \mathbf{z})$ be independent of $\boldsymbol{\varepsilon}$. The incorporation of both forms of heterogeneity then results in the general class of EASI cost functions

$$(B3) \quad C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = \bar{C}(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) + T(\mathbf{p}, \mathbf{z}) + S(\mathbf{p}, \mathbf{z})u,$$

where $\bar{C}$ is a regular ESI cost function and $\exp(T(\mathbf{p}, \mathbf{z}))$ and $\exp(S(\mathbf{p}, \mathbf{z}))$ are both homogeneous of degree zero in $\exp(\mathbf{p})$. Based on equation (B2) and Theorems 3 and 4, one very flexible choice for $\bar{C}$ is $\bar{C}(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = u + \mathbf{p}'\mathbf{m}(u, \mathbf{z}, \boldsymbol{\varepsilon})$, which allows for nonseparability with respect to the unobserved heterogeneity parameters $\boldsymbol{\varepsilon}$, and results in an EASI form

$$(B4) \quad C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = u + \mathbf{p}'\mathbf{m}(u, \mathbf{z}, \boldsymbol{\varepsilon}) + T(\mathbf{p}, \mathbf{z}) + S(\mathbf{p}, \mathbf{z})u.$$

We will use this specification for illustration in Theorems 6 through 8 in Appendix C and in the online Appendix.

Replacing $\mathbf{m}(u, \mathbf{z}, \boldsymbol{\varepsilon})$ with $\mathbf{m}(u, \mathbf{z}) + \boldsymbol{\varepsilon}$ in equation (B4) yields Hicksian (and hence implicit Marshallian) budget shares that are additively separable in $\boldsymbol{\varepsilon}$, which is convenient for estimation. This results in the EASI cost function form

$$(B5) \quad C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = u + \mathbf{p}'\mathbf{m}(u, \mathbf{z}) + T(\mathbf{p}, \mathbf{z}) + S(\mathbf{p}, \mathbf{z})u + \mathbf{p}'\boldsymbol{\varepsilon},$$

where $\mathbf{1}'\mathbf{m}(u, \mathbf{z}) = 1$ and $\mathbf{1}'\boldsymbol{\varepsilon} = 0$. We use this form as the basis for our empirical investigation and for most discussion in the text.

APPENDIX C: GLOBAL REGULARITY OF EASI DEMANDS

THEOREM 6: Assume a log cost function in the general EASI form of equation (B4). Sufficient conditions for the cost function to be regular everywhere on the support of the data are $\mathbf{1}'\mathbf{m}(u, \mathbf{z}, \boldsymbol{\varepsilon}) = 1$, $\exp[T(\mathbf{p}, \mathbf{z})]$ and $\exp[S(\mathbf{p}, \mathbf{z})]$ homogeneous of degree zero in $\exp(\mathbf{p})$, $T(\mathbf{p}, \mathbf{z})$ and $S(\mathbf{p}, \mathbf{z})$ concave and differentiable in $\mathbf{p}$, $1 + \inf[\mathbf{p}'\nabla_u\mathbf{m}(u, \mathbf{z}, \boldsymbol{\varepsilon})] > \inf[S(\mathbf{p}, \mathbf{z})]$ and $\inf(u) > 0$, where these infima are over the supports of $\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}$.

PROOF OF THEOREM 6: Homogeneity of the cost function follows from $\mathbf{1}'\mathbf{m}(u, \mathbf{z}, \boldsymbol{\varepsilon}) = 1$ and homogeneity of $\exp T(\mathbf{p}, \mathbf{z})$ and $\exp S(\mathbf{p}, \mathbf{z})$. Concavity of $T(\mathbf{p}, \mathbf{z})$ and $S(\mathbf{p}, \mathbf{z})$ and having u positive means that $T(\mathbf{p}, \mathbf{z}) + S(\mathbf{p}, \mathbf{z})u$ is concave, which suffices for cost function concavity. For monotonicity $\nabla_u C(\mathbf{p}, u, \mathbf{z}, \boldsymbol{\varepsilon}) = 1 + \mathbf{p}'\nabla_u\mathbf{m}(u, \mathbf{z}, \boldsymbol{\varepsilon}) + S(\mathbf{p}, \mathbf{z})$, which is positive given $1 + \inf[\mathbf{p}'\nabla_u\mathbf{m}(u, \mathbf{z}, \boldsymbol{\varepsilon})] > -\inf[S(\mathbf{p}, \mathbf{z})]$.

By Theorem 6, EASI demands can be globally (i.e., over the entire supports of the data) regular if the functions T and S satisfy homogeneity and concavity conditions, and if T , S , $\mathbf{m}$, and

the supports of the data are suitably bounded. The sufficient, though stronger than necessary, conditions in Theorem 6 include $\inf(u) > 0$ (recall that u here is expressed in the cardinalization that makes $u = y$). This condition can always be satisfied without loss of generality by suitably adjusting the units used to measure expenditures. To see this, define $K_* = \inf[\mathbf{p}'\mathbf{m}(\tilde{u}, \mathbf{z}, \epsilon) + T(\mathbf{p}, \mathbf{z})]$ where this infimum is over all $\mathbf{p}, \mathbf{z}, \epsilon$ on their support, and over all $\tilde{u} \geq 0$. If K_* exists, $\inf[S(\mathbf{p}, \mathbf{z})] > -1$ and $\inf(x) > 0$, then dividing the units expenditures are measured in by $\exp(K_*)$ makes the log of these rescaled expenditures x satisfy $\inf(x) > K_*$. This then suffices to make $\inf(u) > 0$ since, using equation (2), $u = [x - \mathbf{p}'\mathbf{m}(u, \mathbf{z}, \epsilon) + T(\mathbf{p}, \mathbf{z})]/[1 + S(\mathbf{p}, \mathbf{z})] \geq [\inf(x) - K_*]/[1 + \inf(S(\mathbf{p}, \mathbf{z}))] > 0$.

In our empirical model based on the cost function (5), $T(\mathbf{p}, \mathbf{z})$ and $S(\mathbf{p}, \mathbf{z})$ satisfy homogeneity, are bounded given boundedness of prices, and are concave if $\sum_{l=0}^L z_l \mathbf{A}_l$ and $\mathbf{B}$ are negative semidefinite. Let $c_r = \mathbf{p}'\mathbf{b}_r$, so boundedness of the support of $\mathbf{p}$ means that each c_r lies in some interval. For monotonicity in our empirical model, assume y is everywhere positive, let $\lambda_0 = [\inf(x) - \sup(\mathbf{p}'\mathbf{1}_J - \sum_{l=0}^L z_l \mathbf{p}'\mathbf{A}_l \mathbf{p}/2)]/[1 + \sup(-\mathbf{p}'\mathbf{B}\mathbf{p})]$ and let $\lambda_1 = \sup(x)$. It follows from equation (8) that $\lambda_0 \leq y \leq \lambda_1$. Let $\tau = -\inf[1 + \mathbf{p}'\mathbf{D}\mathbf{z} + \mathbf{p}'\mathbf{B}\mathbf{p}/2]$. Since $c_r = \mathbf{p}'\mathbf{b}_r$, boundedness of the support of $\mathbf{p}$ means that each c_r lies in some interval. Then the sufficient monotonicity condition in Theorem 6 is satisfied if $\sum_{r=0}^5 c_r r y^{r-1} > \tau$ holds for all c_r on their supports and all y in the interval $\lambda_0 \leq y \leq \lambda_1$. In particular, if quantities are scaled so $\mathbf{p}$ is nonnegative, and recalling that $\lambda_0 > 0$, a sufficient condition is existence of a finite lower bound on the slope of every Engel curve at every point. More generally, monotonicity requires that downward sloping portions of Engel curves for any good need to be sufficiently offset by upward slopes of other Engel curves.

In practice, imposing the restrictions on parameter values that are sufficient but not necessary to guarantee global regularity of demand systems can induce undesirable restrictions on flexibility. See, e.g., Diewert and Wales (1987) and Ryan and Wales (1998). In our empirical application, we therefore follow the common practice of estimating without imposing the inequality constraints associated with regularity, and afterward check for regularity in the neighborhood of the data.

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